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AXI Multichannel DMA IP for PCI Express* User Guide

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1. About This Document

This document provides information on the Quartus® Prime Software-generated AXI Multichannel DMA IP for PCI Express for the Agilex™ 7 I-Series FPGAs with R-Tile variant.

The AXI Multichannel DMA IP for PCI Express is integrated with the AXI Streaming IP for PCI Express to provide high performance PCI Express connectivity.

Note: This IP inherits most of the capabilities provided by the legacy Avalon MCDMA IP. For information about this Avalon MCDMA IP, refer to the Multi Channel DMA IP for PCI Express User Guide.

1.1. Intended Audience

This document is intended for FPGA and system developers to use as a guide for evaluating the AXI Multichannel DMA FPGA IP for PCI Express in conjunction with AXI Streaming FPGA IP for PCI Express on Agilex 7 I-Series FPGA device.

1.2. Conventions

Table 1. Conventions

Convention	Description
#	Precedes a command and indicates the command is to be entered as root.
$	Indicates a command is to be entered as a user.
`This font`	Filenames, commands, and keywords are printed in this font. Long command lines are printed in this font. Although long command lines may wrap to the next line, the return is not part of the command. Do not press enter.
<variable_name>	Indicates the placeholder text that appears between the angle brackets must be replaced with an appropriate value. Do not enter the angle brackets.

1.3. Terms and Acronyms

Table 2. Glossary of Terms

Term	Description
AXI	Advanced eXtensible Interface
AXI-MM	AXI Memory-Mapped
AXI-ST	AXI Stream

continued...

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Term	Description
AXI MCDMA IP	Abbreviation used in this document. Refers to the *AXI Multichannel DMA FPGA IP for PCI Express*
AXI Streaming IP	Abbreviation used in this document. Refers to the *AXI Streaming FPGA IP for PCI Express*
BAM	Bursting Master
BAS	Bursting Slave
Channel	A DMA channel consists of a pair of Host-to-Device (H2D) and Device-to-Host (D2H) descriptor queues to handle bidirectional data transfer
CSR	Control and Status Register
DMA	Direct Memory Access
D2H	Device-to-Host
Gen1	PCIe 1.0
Gen2	PCIe 2.0
Gen3	PCIe 3.0
Gen4	PCIe 4.0
Gen5	PCIe 5.0
H2D	Host-to-Device
HIP	Hard IP
Avalon MCDMA IP	H/P/F/R-Tile Multichannel DMA FPGA IP for PCI Express. These IPs use traditional Avalon® interface protocol
MCDMA	Multichannel Direct Memory Access
QCSR	Queue Control and Status Register
TLP	Transaction Layer Packet

1.4. Related Information

Multi Channel DMA FPGA IP for PCI Express User Guide

Multi Channel DMA FPGA IP for PCI Express Design Example User Guide

AXI Streaming FPGA IP for PCI Express User Guide

AMBA AXI4-Stream Protocol Specification

AMBA AXI Protocol Specification

PCI Express Base Specification Revision 5.0

Agilex 7 FPGA I-Series Development Kit

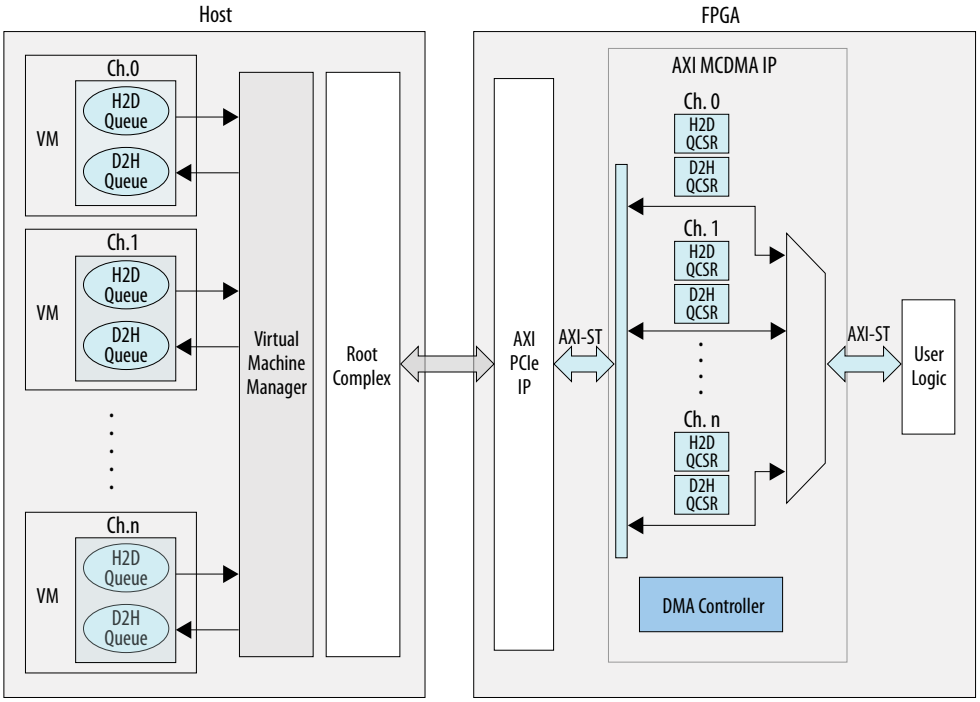
2. Introduction

2.1. Overview

The AXI Multichannel DMA IP for PCI Express* (AXI MCDMA IP) allows you to implement Multichannel DMA using industry standard AXI interface protocol. The AXI MCDMA IP interfaces with AXI Streaming IP for PCI Express (AXI-ST PCIe IP). It facilitates streamlined and high-performance data transfer by furnishing independent DMA channels that operate seamlessly over the underlying PCIe link connecting the host and the device.

The figure below shown an example DMA application where the AXI MCDMA IP plays a crucial role in the hardware infrastructure of a server, enabling smooth communication between various VM clients and their FPGA-device counterparts. The AXI MCDMA IP operates on descriptor-based queues (there is a maximum of 65,535 descriptors per queue), established by the driver software, facilitating efficient data transfers between the local FPGA and the host. The control logic embedded in the AXI MCDMA IP for PCI Express intelligently interprets and executes the queued descriptors, ensuring the reliability and efficiency of data transfer operations.

Figure 1. AXI MCDMA IP for PCI Express : Server Hardware Infrastructure Use Case Example



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2.2. IP Features

The AXI MCDMA IP includes the functionality of the Avalon MCDMA IP, with the primary difference lying in the added support for PCIe Gen5 1x16 and transitioning from Avalon to AXI interface protocol.

AXI MCDMA IP Features

- Interfaces with AXI Streaming FPGA IP for PCI Express in Agilex 7 I-Series with R-Tile variants.
- Supports PCIe Gen3/4/5 1x16, 1x8, and 1x4 Root Port and Endpoint modes.

  Note: ☐ To generate 1x8, or 1x4 IP, create a `quartus.ini` file in the project directory with the content: `ini_guard=on`.

  ☐ 2x8 mode is supported for the design example only.

  ☐ When the AXI MCDMA IP in Gen3 1x4 mode connects to the AXI Streaming IP in Gen4 x2 mode with a 128b interface, the link comes up as a Gen4 x2 link.
- User interface bandwidth: 256/512/1024-bits @ up to 500 MHz

2.2.1. Root Port Mode Features

- Address Translation Table (ATT) support in BAS and BAM+BAS mode
- Automatically enables Configuration Slave and Hard IP Reconfiguration interfaces
- 10-bit tag support
- Completion reordering
- Completion timeout
- User mode and supported user interface
  - ☐ Bursting Master
    - BAM AXI-MM master
    - Configuration Slave
    - Hard IP Reconfiguration
  - ☐ Bursting Slave
    - BAS AXI-MM slave
    - Configuration Slave
    - Hard IP Reconfiguration
  - ☐ BAM + BAS
    - BAM AXI-MM master
    - BAS AXI-MM slave
    - Configuration Slave
    - Hard IP Reconfiguration

2.2.2. Endpoint Mode Features

- Supports 2048 DMA channels (Maximum 512 channels per Function)
- H2D/D2H DMA data transfer via AXI-ST or AXI-MM interface
- Integrated MSI-X for DMA operation and User MSI-X for user applications
- SR-IOV support with 8 PFs and 2048 VFs
- 10-bit tag support
- Completion reordering
- Completion timeout
- User Function Level Reset

- 64-bit Metadata
- User mode and supported user interfaces
  - ☐ Multichannel DMA
    - H2D AXI-ST source
    - D2H AXI-ST sink
    - H2D/D2H AXI-MM master
    - PIO AXI-Lite master
    - User MSI-X
    - User FLR
  - ☐ Bursting Master
    - BAM AXI-MM master
  - ☐ Bursting Slave
    - BAM AXI-MM slave
    - User MSI
  - ☐ BAM + BAS
    - BAM AXI-MM master
    - BAS AXI-MM slave
    - User MSI
  - ☐ BAM + MCDMA
    - BAM AXI-MM master
    - H2D AXI-ST source
    - D2H AXI-ST sink
    - H2D/D2H AXI-MM master
    - User FLR
    - User MSI-X
  - ☐ BAM + BAS + MCDMA
    - BAM AXI-MM master
    - BAS AXI-MM slave
    - H2D AXI-ST source
    - D2H AXI-ST sink
    - H2D/D2H AXI-MM master
    - User FLR
    - User MSI-X
- Optional User Interfaces
  - ☐ Hard IP Reconfiguration interface
  - ☐ User Configuration Intercept interface

2.2.3. Design Example Support

The table below lists the AXI MCDMA design examples available in the AXI MCDMA IP Parameter Editor under the **Example Designs** tab. Based on the IP parameter settings, such as User Mode and User Interface, you can select a design example, compile it, and perform a hardware test.

Note: Design example simulation may be supported in a future release.

Table 3. AXI MCDMA Endpoint Design Example Support

User Mode	DMA User Interface	Available Design Example
Multichannel DMA	AXI Streaming	AXI-S Device-side Packet Loopback AXI-S Packet Generate/Check
BAM + MCDMA BAM + BAS + MCDMA	AXI Streaming	AXI-S Device-side Packet Loopback
Multichannel DMA Bam + MCDMA BAM + BAS + MCDMA	AXI Memory-Mapped	AXI-MM DMA
BAM + BAS	n/a	AXI-MM Traffic Generator/Checker
Bursting Master (BAM)	n/a	AXI-MM BAM EP Memory

2.3. Device Family Support

The following terms define the device support levels for Altera FPGA IP cores:

- Advanced support: The IP core is available for simulation and compilation for this device family. Timing models include initial engineering estimates of delays based on early post-layout information. The timing models are subject to change as silicon testing improves the correlation between the actual silicon and the timing models. You can use this IP core for system architecture and resource utilization studies, simulation, pinout, system latency assessments, basic timing assessments (pipeline budgeting), and I/O transfer strategy (data-path width, burst depth, I/O standards tradeoffs).
- Preliminary support: The IP core is verified with preliminary timing models for this device family. The IP core meets all functional requirements, but might still be undergoing timing analysis for the device family. It can be used in production designs with caution.
- Final support: The IP core is verified with final timing models for this device family. The IP core meets all functional and timing requirements for the device family and can be used in production designs.

Table 4. Device Family Support

Device Family	Support Level
Agilex 7 I-Series	Final Support

2.4. IP and Design Example Support

Table 5. Endpoint Mode

PCIe Mode	IP Support	Design Example Support	Note
Gen5/4/3 1x16	Yes	Yes	To generate the 1x16 IP or design example: • Select IP PCIe Mode: 1x16. • Select IP Data Width. • Select the design example **PCIe Mode** in the **PCIe Settings** tab of the **Example Designs** tab: 1x16.
Gen5/4/3 1x8	Yes	No	To generate the 1x8 IP: • Create a `quartus.ini` file with content: `ini_guard=on`. • Select IP PCIe Mode: 1x8. • Select IP Data Width. • Select the design example **PCIe Mode** in the **PCIe Settings** tab of the **Example Designs** tab: x8.
Gen5/4/3 1x4	Yes	No	To generate the 1x4 IP: • Create a `quartus.ini` file with content: `ini_guard=on`. • Select IP PCIe Mode: 1x4. • Select IP Data Width. • Select the design example **PCIe Mode** in the **PCIe Settings** tab of the **Example Designs** tab: x4.

Table 6. Root Port Mode

PCIe Mode	IP Support	Design Example Support	Note
Gen5/4/3 1x16	Yes	No	To generate the 1x16 IP: • Select IP PCIe Mode: 1x16. • Select IP Data Width. • Select the design example **PCIe Mode** in the **PCIe Settings** tab of the **Example Designs** tab: x16.
Gen5/4/3 1x8	Yes	No	To generate the 1x8 IP: • Create a `quartus.ini` file with the content: `ini_guard=on`. • Select IP PCIe Mode: 1x8. • Select IP Data Width. • Select the design example **PCIe Mode** in the **PCIe Settings** tab of the **Example Designs** tab: x8.
Gen5/4/3 1x4	Yes	No	To generate the 1x4 IP: • Create a `quartus.ini` file with the content: `ini_guard=on`. • Select IP PCIe Mode: 1x4. • Select IP Data Width. • Select the design example **PCIe Mode** in the **PCIe Settings** tab of the **Example Designs** tab: x4.

Note: For 2x8, 2x4, and 4x4, you can generate the IP as 1x8 or 1x4 and use multiple instances with the appropriate AXI Streaming IP to build design examples.

2.5. Recommended Speed Grade

Table 7. Recommended Speed Grades

PCIe Generation Support	Device	Tile Variant	PLD Clock Frequency	Recommended Fabric Speed Grade
Gen3	Agilex 7	R-Tile	250 MHz	-1, -2, -3
Gen4	Agilex 7	R-Tile	400 MHz	-1, -2, -3
			500 MHz	-1
Gen5	Agilex 7	R-Tile	500 MHz	-1

2.6. Resource Utilization

Table 8. PCIe Gen5 Endpoint Configuration

User Mode	Link Width	DMA Channels	ALMs	Logic Registers	M20Ks
Multichannel DMA	1x16	4	42998.6	146957	820
BAM+BAS	1x16	NA	34343.5	99236	835
BAM+BAS +MCDMA	1x16	4	60867.1	201204	1248

2.7. Release Information

IP versions are the same as the Quartus Prime Design Suite software versions up to v19.1. From Quartus Prime Design Suite software version 19.2 or later, IP cores have a new IP versioning scheme. If an IP core version is not listed, the user guide for the previous IP core version applies. The IP versioning scheme (X.Y.Z) number changes from one software version to another.

A change in:

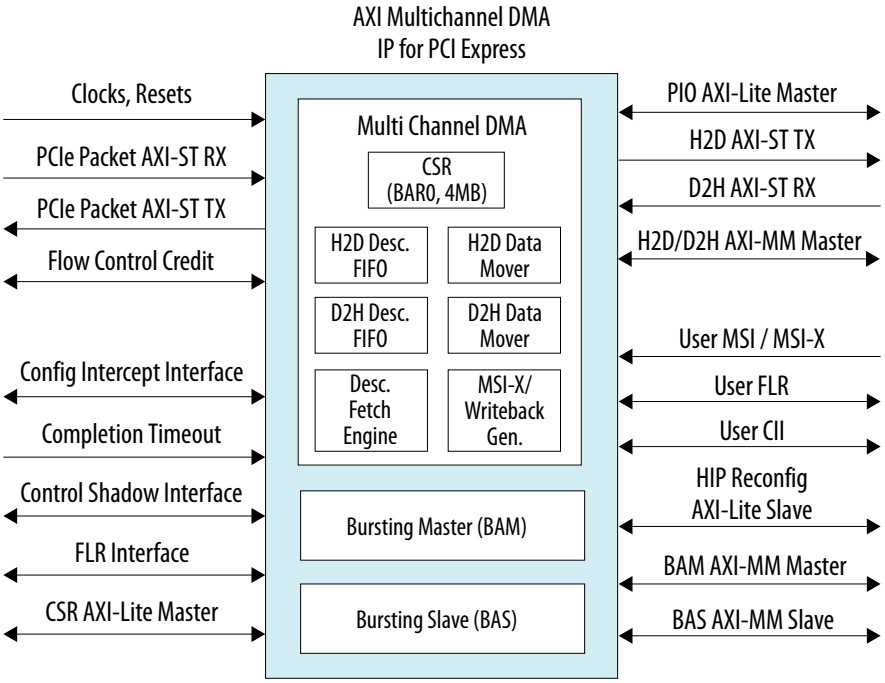
- X indicates a major revision of the IP. If you update your Quartus Prime software, you must regenerate the IP.
- Y indicates the IP includes new features. Regenerate your IP to include these new features.
- Z indicates the IP includes minor changes. Regenerate your IP to include these changes.

Table 9. Release Information

Item	Description
IP Version	3.3.0
Quartus Prime Version	Quartus Prime Pro Edition Software 25.3.1 Release
Ordering Part Number [OPN]	IP-PCIEMCDMA-AXI

3. Functional Description

Figure 2. AXI MCDMA IP Block Diagram



The AXI MCDMA IP inherits most of the legacy Avalon MCDMA IP functionality. For the functional description of this Avalon MCDMA IP, refer to the Multi Channel DMA IP for PCI Express User Guide.

Related Information

Multi Channel DMA IP for PCI Express User Guide

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4. Parameters

This chapter provides a reference for all the IP parameters of the AXI Multichannel DMA IP for PCI Express.

4.1. IP Settings

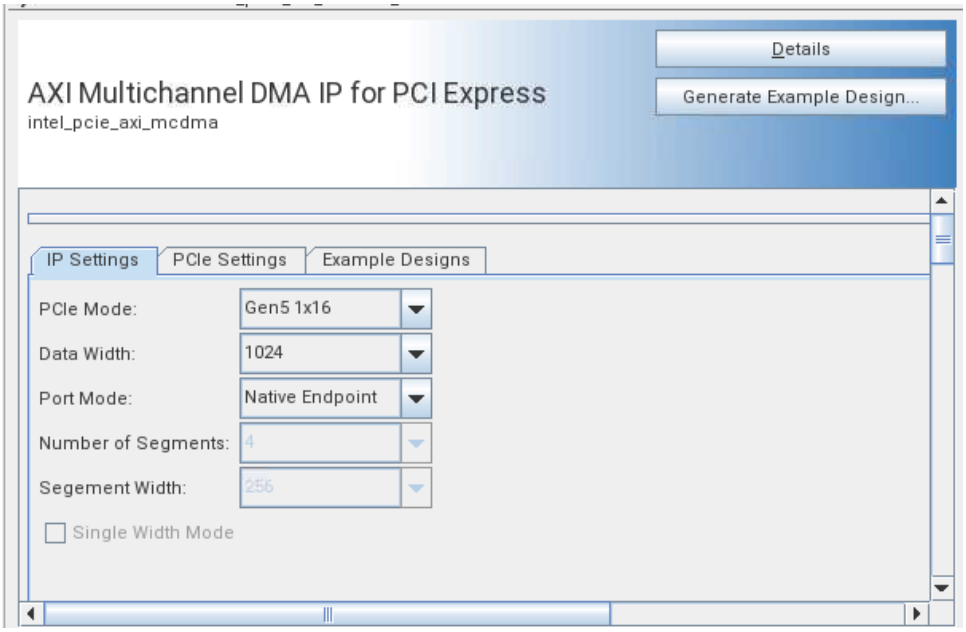

Table 10. IP Settings Parameters

Parameter	Value	Default Value	Description
PCIe Mode	**Gen5 1x16** **Gen4 1x16** **Gen3 1x16** **Gen5 1x8** **Gen4 1x8** **Gen3 1x8** **Gen5 1x4** **Gen4 1x4** **Gen3 1x4**	**Gen5 1x16**	Selects the width of the data interface between the transaction layer and the application layer implemented in the PLD fabric, the lane data rate and the lane rate.

continued...

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Parameter	Value	Default Value	Description
			Note: • The AXI MCDMA IP Parameter Editor does not provide a separate interface for the bifurcated ports. Identical IP configuration settings are used. • To select a bifurcated 2x8 or 4x4 mode, go to the **Example Designs** tab > **PCIe Interfaces** > **PCIe Interface Settings** > **PCIe Mode**. • 1x4 and 1x8 PCIe interface modes for the design example are not supported in the current release.
Data Width	**1024** **512** **256** **128**	**1024**	Supported data widths per the PCIe mode: • Gen5 1x16: 1024 bits • Gen4 1x16: 512 bits • Gen3 1x16: 512 bits • Gen5 1x8: 512 bits • Gen4 1x8: 256 bits • Gen3 1x8: 256 bits • Gen5 1x4: 256 bits • Gen4 1x4: 256 bits, 128 bits • Gen3 1x4: 256 bits, 128 bits
Port Mode	**Native Endpoint** **Root Port**	**Native Endpoint**	Selects port mode.
Number of Segments	**4** **2** **1**	**4**	Number of segments in data interface. IP Parameter Editor automatically selects a value per the Data Width: • 1024 bits: 4 • 512 bits: 2 • 256 bits: 2 • 128 bits: 1
Segment Width	**256** **128**	**256**	Segment data width. Fixed at 256 bits for x16 and x8 modes, and 128 bits for x4 mode.
Single Width Mode		**Off**	Sets the Single Width Mode. IP Parameter Editor automatically sets this to **On** based on the **PCIe Mode** and **Data Width** settings.

4.1.1. PCIe Settings

4.1.1.1. MCDMA Settings

4.1.1.1.1. Multichannel DMA Mode

Figure 3. Multichannel DMA Mode Parameters

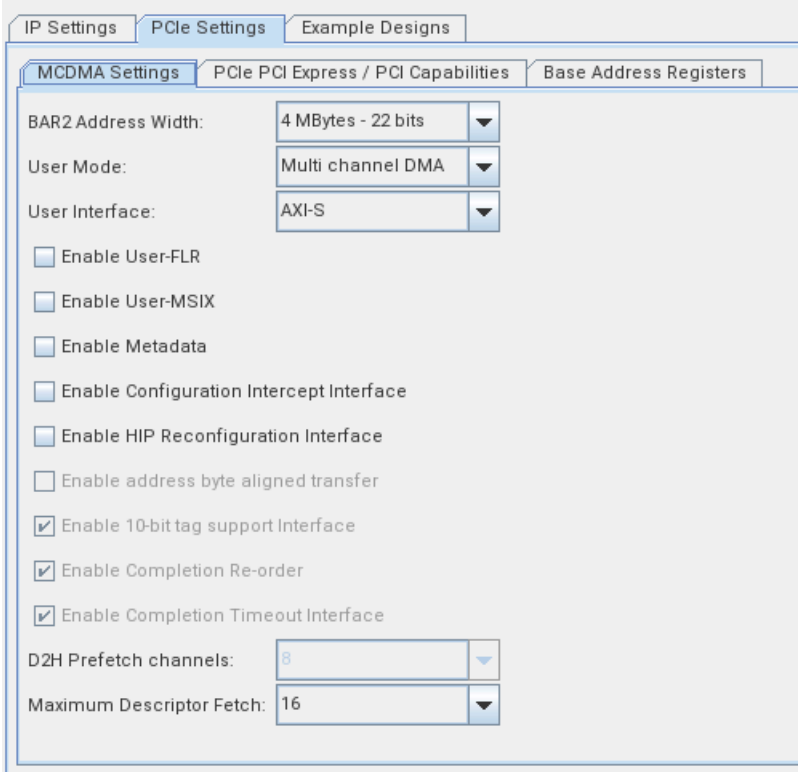


Table 11. Multichannel DMA Mode Parameters

Parameter	Value	Default Value	Description
BAR2 Address Width	**128 Bytes – 7 bits** **. . .** **8 EBytes – 63 bits**	**4 Mbytes – 22 bits**	Sets the BAR2 size used for PIO functionality. *Note:* This parameter is available only in **Multichannel DMA** user mode.
User Mode	**Multichannel DMA** **Bursting Master** **Bursting Slave** **BAM+BAS** **BAM+MCDMA** **BAM+BAS+MCDMA**	**Multichannel DMA**	This parameter allows you to configure the mode of operation for the user application: • Bursting Master (BAM) and Bursting Slave (BAS) offer bursting AXI-MM capabilities without DMA functionality
User Interface	**AXI-S** **AXI-MM**	**AXI-S**	Sets the type of DMA user interface. You can select either the AXI Stream or AXI Memory-Mapped interface for DMA.

continued...

Parameter	Value	Default Value	Description
Enable User-FLR	**On / Off**	**Off**	Select to enable User FLR interface which allows passing of FLR signals to the user side application. *Note:* User FLR is not supported in Bursting Master, Bursting Slave, and BAM+BAS modes.
Enable User-MSIX	**On / Off**	**Off**	User MSI-X interface enables the user application to initiate interrupts through the MCDMA IP. *Note:* User MSI-X is not supported in Bursting Master, Bursting Slave, and BAM+BAS modes.
Enable Metadata	**On / Off**	**Off**	Enables or disables 8-byte metadata. Metadata is the 8-byte user application data that is transferred to and from the device via the DEST address field (for H2D DMA) and SRC address field (for D2H DMA) in the AXI Streaming mode. *Note:* Metadata is not supported in Bursting Master, Bursting Slave, and BAM+BAS modes.
Enable Configuration Intercept Interface	**On / Off**	**Off**	Select to enable user configuration intercept interface.
Enable HIP Reconfiguration Interface	**On / Off**	**Off**	Select to enable Hard IP reconfiguration interface.
Enable address byte aligned transfer	**Off**	**Off**	This option is not supported in the current release.
Enable 10-bit tag support interface	**On**	**On**	Enables 10-bit tag support interface
Enable Completion Re-order	**On**	**On**	Enables completion re-order
Enable Completion Timeout Interface	**On**	**On**	Enables completion timeout interface
D2H Prefetch channels	**8** **16** **32** **64** **128** **256** **512**	**8**	Sets the number of D2H prefetch channels for the AXI-S interface type. IP Parameter Editor automatically selects the value based on the DMA channel count.
Maximum Descriptor Fetch	**16** **32** **64**	**16**	Sets the maximum descriptors that are fetched per D2H prefetch channel. Applicable to the AXI-S interface type.

4.1.1.1.2. Bursting Master Mode

Figure 4. Root Port Bursting Master Mode Parameters

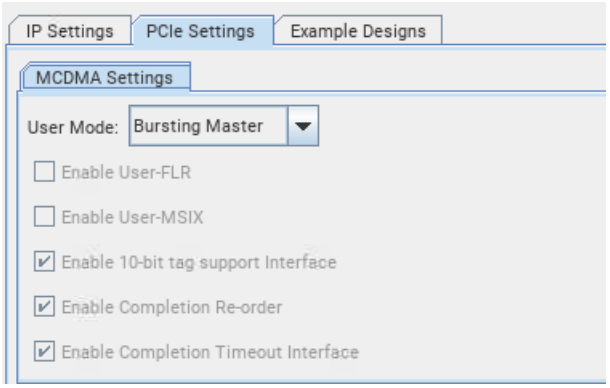


Figure 5. Endpoint Bursting Master Mode Parameters

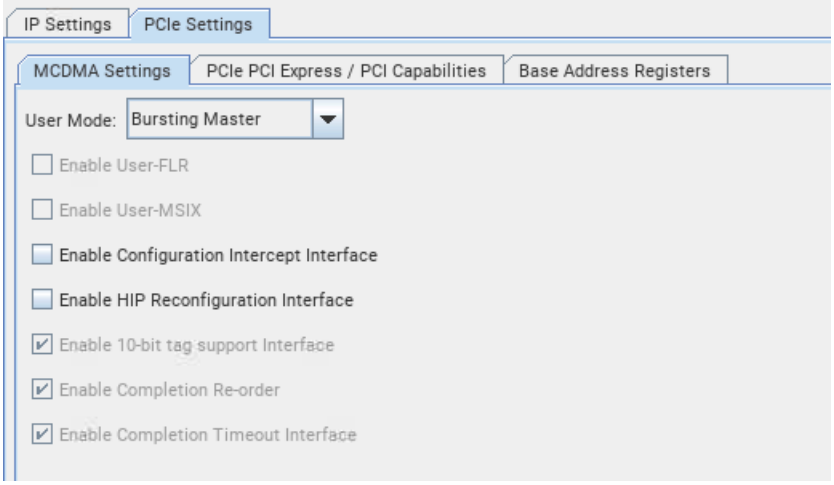


Table 12. Bursting Master Mode Parameters

Parameter	Value	Default Value	Description
Enable Configuration Intercept Interface	**On / Off**	**Off**	Select to enable user configuration intercept interface. Applicable to Endpoint only.
Enable HIP Reconfiguration Interface	**On / Off**	**Off**	Select to enable Hard IP reconfiguration interface. Applicable to Endpoint only.
Enable 10-bit tag support interface	**On**	**On**	Enables 10-bit tag support interface.
Enable Completion Re-order	**On**	**On**	Enables completion re-order.
Enable Completion Timeout Interface	**On**	**On**	Enables completion timeout interface.

4.1.1.1.3. Bursting Slave Mode

Figure 6. Root Port Bursting Slave Parameters

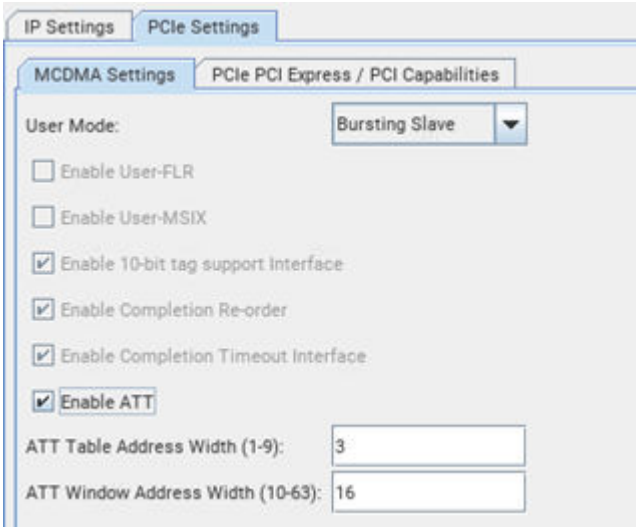


Table 13. Root Port Bursting Slave Parameters

Parameter	Value	Default Value	Description
Enable Configuration Intercept Interface	**On / Off**	**Off**	Select to enable user configuration intercept interface.
Enable HIP Reconfiguration Interface	**On / Off**	**Off**	Select to enable Hard IP reconfiguration interface.
Enable 10-bit tag support interface	**On**	**On**	Enables 10-bit tag support interface
Enable Completion Re-order	**On**	**On**	Enables completion re-order
Enable Completion Timeout Interface	**On**	**On**	Enables completion timeout interface
Enable ATT	**On / Off**	**Off**	Enables Address Translation Table for BAS. *Note:* ATT is supported only in the Root Port mode Bursting Slave.
ATT Table Address Width	**1 - 9**	**3**	Sets depth of ATT. Depth is equal to 2 to the power of number entered.
ATT Window Address Width	**10 - 63**	**16**	Sets the number of BAS address bits to be used directly.

Figure 7. Endpoint Bursting Slave Parameters

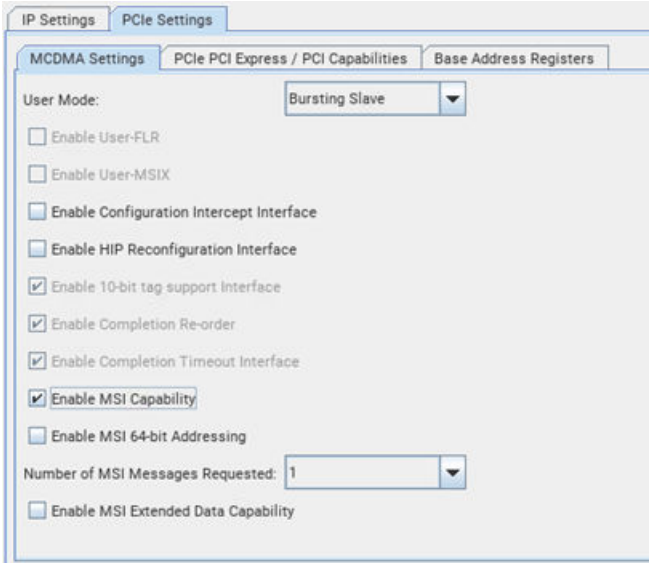


Table 14. Endpoint Bursting Slave Parameters

Parameter	Value	Default Value	Description
Enable Configuration Intercept Interface	**On / Off**	**Off**	Select to enable user configuration intercept interface.
Enable HIP Reconfiguration Interface	**On / Off**	**Off**	Select to enable Hard IP reconfiguration interface.
Enable 10-bit tag support interface	**On**	**On**	Enables 10-bit tag support interface.
Enable Completion Re-order	**On**	**On**	Enables completion re-order.
Enable Completion Timeout Interface	**On**	**On**	Enables completion timeout interface.
Enable MSI Capability	**On / Off**	**Off**	Enables or disables MSI capability. *Note:* MSI is supported only in BAS and BAM + BAS modes.
Enable MSI 64-bit Addressing	**On / Off**	**Off**	Enables or disables 64-bit MSI addressing.
Number of MSI Messages Requested	**1, 2, 4, 8, 16, 32**	**1**	Sets the number of messages that the application can request in the multiple message capable field of the Message Control register.
Enable MSI Extended Data Capability	**On / Off**	**Off**	Enables or disables MSI extended data capability.

4.1.1.1.4. BAM+BAS Mode

Table 15. Root Port Bursting Slave Parameters

Parameter	Value	Default Value	Description
Enable Configuration Intercept Interface	**On / Off**	**Off**	Select to enable user configuration intercept interface.
Enable HIP Reconfiguration Interface	**On / Off**	**Off**	Select to enable Hard IP reconfiguration interface.
Enable 10-bit tag support interface	**On**	**On**	Enables 10-bit tag support interface
Enable Completion Re-order	**On**	**On**	Enables completion re-order
Enable Completion Timeout Interface	**On**	**On**	Enables completion timeout interface
Enable ATT	**On / Off**	**Off**	Enables Address Translation Table for BAS. *Note:* ATT is supported only in the Root Port mode Bursting Slave.
ATT Table Address Width	**1 - 9**	**3**	Sets depth of ATT. Depth is equal to 2 to the power of number entered.
ATT Window Address Width	**10 - 63**	**16**	Sets the number of BAS address bits to be used directly.

Figure 8. Endpoint BAM+BAS Mode Parameters

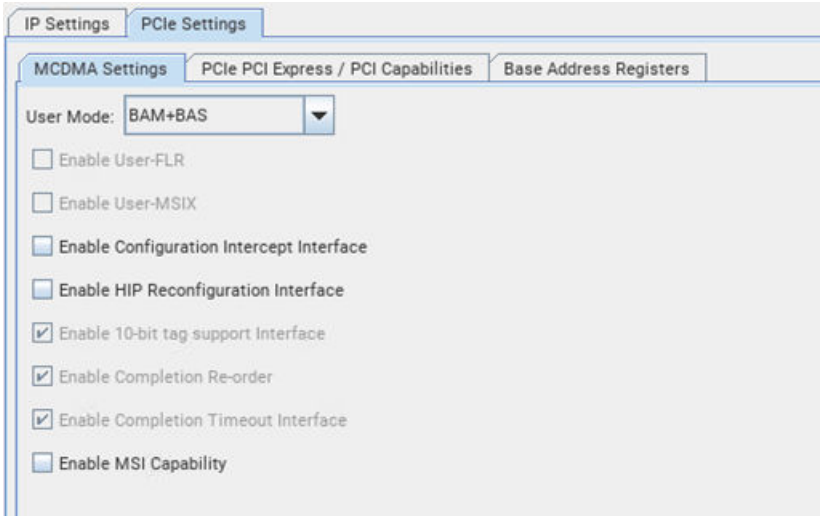


Table 16. Endpoint BAM+BAS Mode Parameters

Parameter	Value	Default Value	Description
Enable Configuration Intercept Interface	**On / Off**	**Off**	Select to enable user configuration intercept interface.
Enable HIP Reconfiguration Interface	**On / Off**	**Off**	Select to enable Hard IP reconfiguration interface.
Enable 10-bit tag support interface	**On**	**On**	Enables 10-bit tag support interface
Enable Completion Re-order	**On**	**On**	Enables completion re-order
Enable Completion Timeout Interface	**On**	**On**	Enables completion timeout interface
Enable MSI Capability	**On / Off**	**Off**	Enables or disables MSI capability *Note:* MSI is supported only in BAS and BAM+BAS modes.

4.1.1.1.5. BAM+MCDMA Mode

Figure 9. BAM+MCDMA Mode Parameters

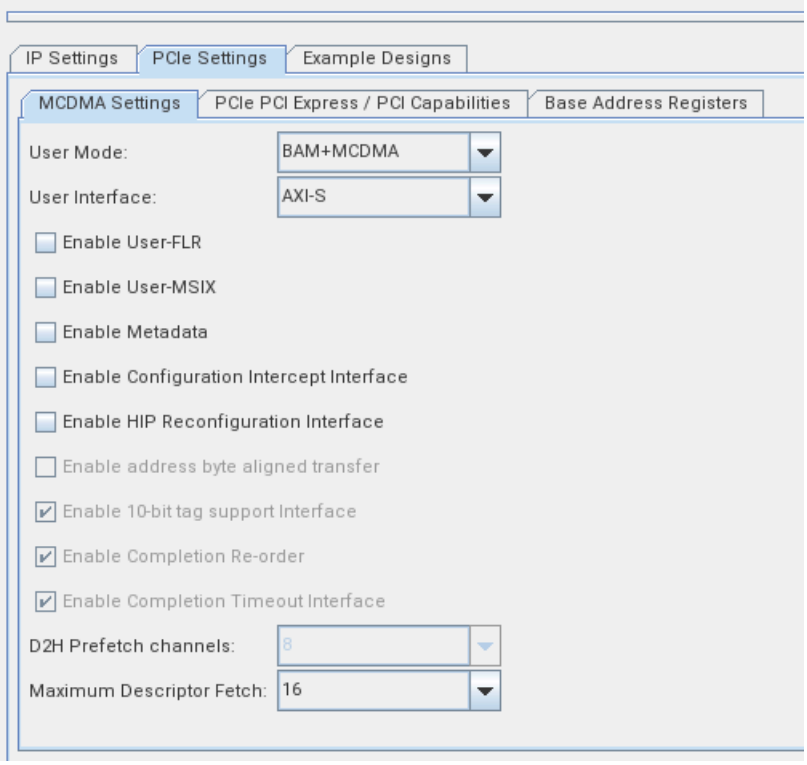


Table 17. BAM+MCDMA Mode Parameters

Parameter	Value	Default Value	Description
User Interface	**AXI-S**	**AXI-S**	Sets the type of DMA User interface.

continued...

Parameter	Value	Default Value	Description
	AXI-MM		You can select either the AXI Stream or AXI Memory-Mapped interface for DMA.
Enable User-FLR	**On / Off**	**Off**	Select to enable User FLR interface which allows the passing of FLR signals to the user side application. *Note:* User FLR is not supported in Bursting Master, Bursting Slave, and BAM + BAS modes.
Enable User-MSIX	**On / Off**	**Off**	User MSI-X interface enables the user application to initiate interrupts through the MCDMA IP. *Note:* User MSI-X is not supported in the Bursting Master, Bursting Slave, and BAM + BAS modes.
Enable Metadata	**On / Off**	**Off**	Enables or disables 8-byte metadata. Metadata is the 8-byte user application data that is transferred to and from the device via the DEST address field (for H2D DMA) and SRC address field (for D2H DMA) in the AXI Streaming mode.
Enable Configuration Intercept Interface	**On / Off**	**Off**	Select to enable user configuration intercept interface.
Enable HIP Reconfiguration Interface	**On / Off**	**Off**	Select to enable Hard IP reconfiguration interface.
Enable address byte aligned transfer	**Off**	**Off**	This option is not supported in the current release.
Enable 10-bit tag support interface	**On**	**On**	Enables 10-bit tag support interface
Enable Completion Re-order	**On**	**On**	Enables completion re-order
Enable Completion Timeout Interface	**On**	**On**	Enables completion timeout interface

4.1.1.1.6. BAM+BAS+MCDMA Mode

Figure 10. BAM+BAS+MCDMA Mode Parameters

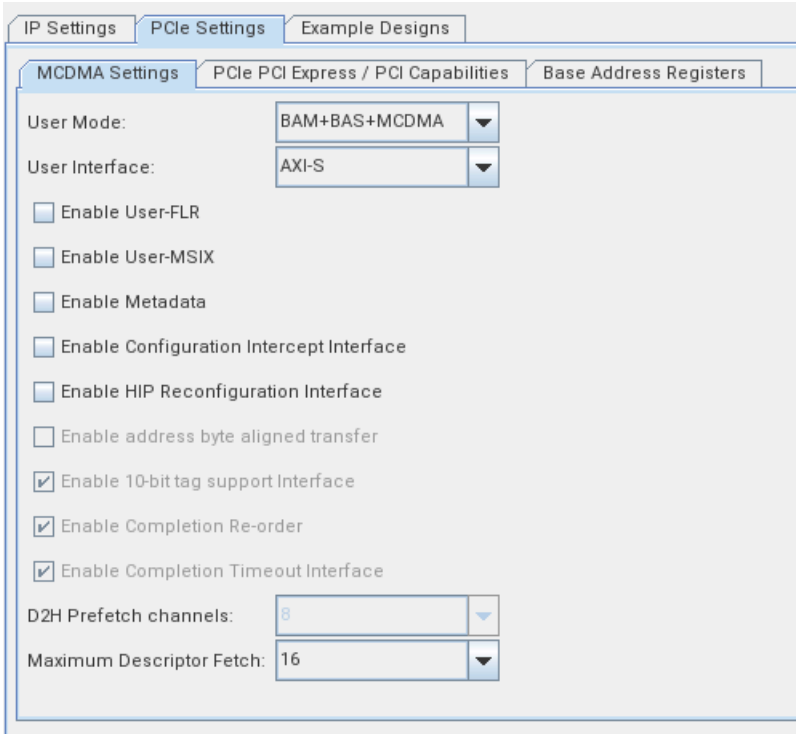


Table 18. BAM+BAS+MCDMA Mode Parameters

Parameter	Value	Default Value	Description
User Interface	**AXI-S** **AXI-MM**	**AXI-S**	Sets the type of DMA user interface. You can select either the AXI Stream or AXI Memory-Mapped interface for DMA.
Enable User-FLR	**On / Off**	**Off**	Select to enable the User FLR interface, which allows the passing of FLR signals to the user side application. *Note:* User FLR is not supported in Bursting Master, Bursting Slave, and BAM + BAS modes.
Enable User-MSIX	**On / Off**	**Off**	User MSI-X interface enables the user application to initiate interrupts through the MCDMA IP. *Note:* User MSI-X is not supported in Bursting Master, Bursting Slave, and BAM + BAS modes.
Enable Metadata	**On / Off**	**Off**	Enables or disables 8-byte metadata. Metadata is the 8-byte user application data that is transferred to and from the device via the DEST address field (for H2D DMA) and SRC address field (for D2H DMA) in the AXI Streaming mode.

continued...

Parameter	Value	Default Value	Description
Enable Configuration Intercept Interface	**On / Off**	**Off**	Select to enable user configuration intercept interface
Enable HIP Reconfiguration Interface	**On / Off**	**Off**	Select to enable Hard IP reconfiguration interface
Enable address byte aligned transfer	**Off**	**Off**	This option is not supported in the current release.
Enable 10-bit tag support interface	**On**	**On**	Enables 10-bit tag support interface
Enable Completion Re-order	**On**	**On**	Enables completion re-order
Enable Completion Timeout Interface	**On**	**On**	Enables completion timeout interface

4.1.1.2. PCI Express / PCI Capabilities

This group of parameters defines various capability properties of the IP core. Some of these parameters are stored in the PCI Configuration Space - PCI Compatible Configuration Space. The byte offset indicates the parameter address.

4.1.1.2.1. PCIe Device

Figure 11. PCIe Multifunction and SR-IOV System Settings Parameters

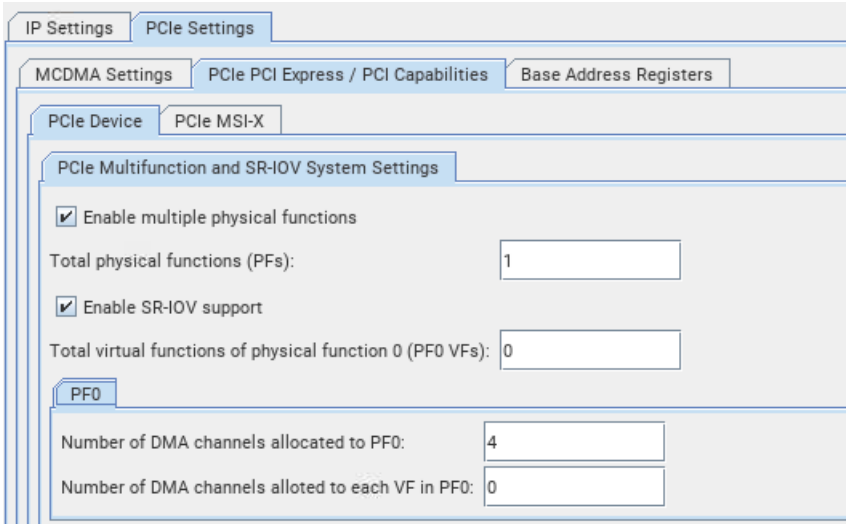


Table 19. PCIe Multifunction and SR-IOV System Settings Parameters

Parameter	Value	Default Value	Description
Enable multiple physical functions	**On / Off**	**Off**	Enables multiple physical functions
Total physical functions (PFs)	**1 – 8**	**1**	Sets the number of physical functions This parameter is available when Enable multiple physical functions is On

continued...

Parameter	Value	Default Value	Description
Enable SR-IOV support	**On / Off**	**Off**	Enable SR-IOV
Total virtual functions of physical function 0 (PF0VFs)	**0 – 2048**	**0**	Sets the number of VFs to be assigned to Physical Function 0 This parameter is available when Enable SR-IOV support is On
Number of DMA channels allocated to PF0	**0 – 512**	**4**	Sets the number of DMA channels allocated to the PF0
Number of DMA channels allocated to each VF in PF0	**0 – 512**	**0**	Sets the number of DMA channels allocated to each VF in PF0

4.1.1.2.2. PCIe MSI-X

PCIe PF MSI-X

The PF MSI-X capability parameters in Multichannel DMA, BAM+MCDMA and BAM+BAS +MCDMA modes are automatically set and cannot be modified.

Figure 12. PCIe PF MSI-X Parameters

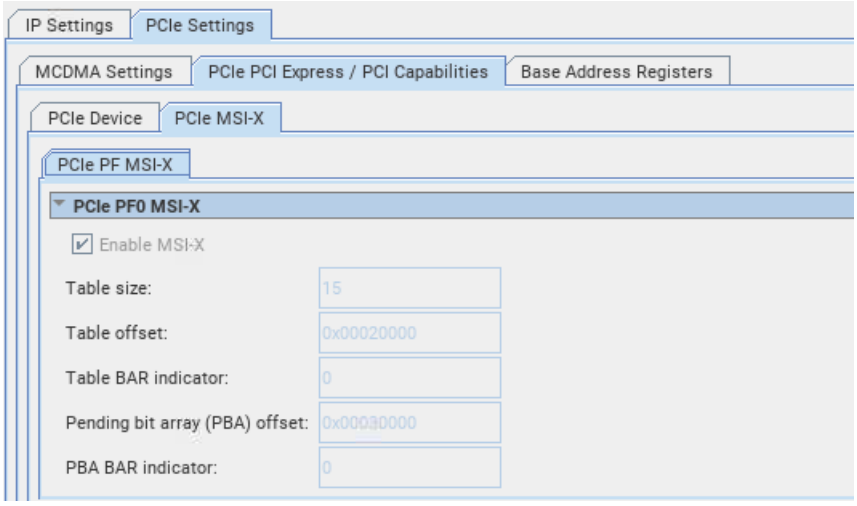


Table 20. PCIe PF MSI-X Parameters

Parameter	Value	Default Value	Description
Enable MSI-X	**On / Off**	See Description	Enables or disables the MSI-X capability. *Note:* In **Multichannel DMA**, **BAM +MCDMA** and **BAM+BAS+MCDMA** user modes, the MSI-X capability is automatically enabled and capability register are set by the IP GUI. You cannot modify the capability register values.
Table Size	**0 - 2047**	**15**	Sets the number of entries in the MSI-X table

continued...

Parameter	Value	Default Value	Description
			System software reads this field to determine the MSI-X table size N, which is encoded as N-1
Table offset	**0x00020000**	**0x00020000**	Sets the read-only base address of the MSI-X table Points to the base of the MSI-X table. The lower 3 bits of the table BAR indicator (BIR) are set to zero by software to form a 64-bit qword-aligned offset
Table BAR Indicator	**0**	**0**	Specifies which one of a function's base address registers, located beginning at 0x10 in the Configuration Space, maps the MSI-X table into memory space This field is read-only
Pending bit array (PBA) offset	**0x00030000**	**0x00030000**	Used as an offset from the address contained in one of the function's Base Address registers to point to the base of the MSI-X PBA The lower 3 bits of the PBA BIR are set to zero by software to form a 32-bit qword-aligned offset. This field is read-only after being programmed
PBA BAR Indicator	**0**	**0**	Specifies the function Base Address registers, located beginning at 0x10 in Configuration Space, that maps the MSI-X PBA into memory space This field is read-only in the MSI-X Capability Structure

PCIe VF MSI-X

The VF MSI-X capability is available in Multichannel DMA, BAM+MCDMA and BAM+BAS +MCDMA user modes if the SR-IOV support is enabled and total VFs are non-zero.

Figure 13. PCIe VF MSI-X Parameters

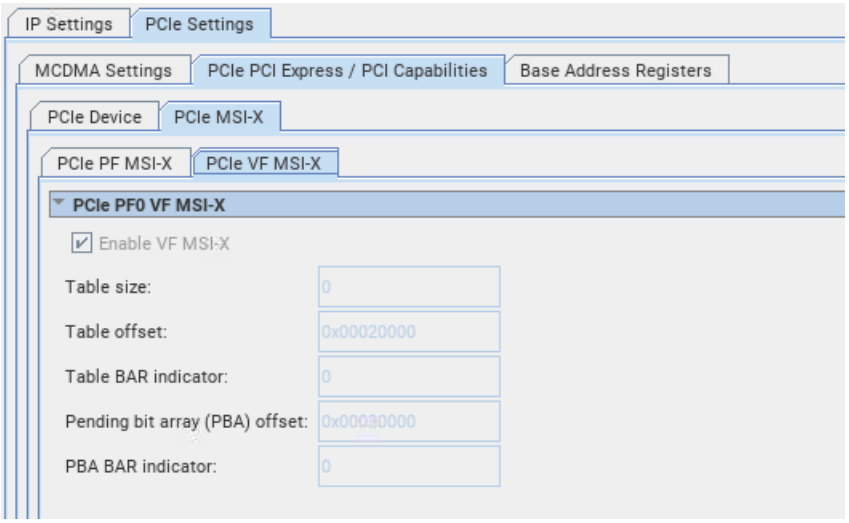
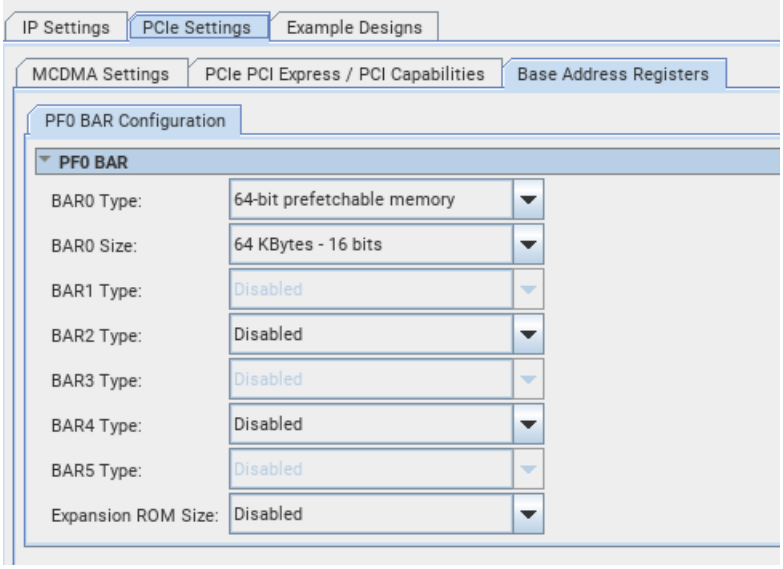


Table 21. PCIe PF MSI-X Parameters

Parameter	Value	Default Value	Description
Enable MSI-X	**On / Off**	See Description	Enables or disables the MSI-X capability *Note:* In **Multichannel DMA**, **BAM +MCDMA** and **BAM+BAS+MCDMA** user modes, the MSI-X capability is automatically enabled and capability register are set by the IP GUI. You cannot modify the capability register values.
Table Size	**0 - 2047**	**0**	Sets the number of entries in the MSI-X table System software reads this field to determine the MSI-X table size N, which is encoded as N-1
Table offset	**0x00020000**	**0x00020000**	Sets the read-only base address of the MSI-X table Points to the base of the MSI-X table. The lower 3 bits of the table BAR indicator (BIR) are set to zero by software to form a 64-bit qword-aligned offset
Table BAR Indicator	**0**	**0**	Specifies which one of a function's base address registers, located beginning at 0x10 in the Configuration Space, maps the MSI-X table into memory space This field is read-only
Pending bit array (PBA) offset	**0x00030000**	**0x00030000**	Used as an offset from the address contained in one of the function's Base Address registers to point to the base of the MSI-X PBA. The lower 3 bits of the PBA BIR are set to zero by software to form a 32-bit qword-aligned offset This field is read-only after being programmed
PBA BAR Indicator	**0**	**0**	Specifies the function Base Address registers, located beginning at 0x10 in Configuration Space, that maps the MSI-X PBA into memory space This field is read-only in the MSI-X Capability Structure

4.1.1.3. Base Address Registers

Figure 14. PF BAR Configuration Parameters



The following table shows the BAR and Expansion ROM parameters that can be configured in each user mode.

In Multichannel DMA mode, BAR2 size can be configured through BAR2 Address Width parameter in MCDMA Settings.

In Multichannel DMA, BAM+MCDMA and BAM+BAS+MCDMA modes, BAR0 is allocated to access 4 MB space that internal IP registers are mapped to.

Supported BAR types for user application are as follows:

- BAR 0/2/4: Disabled, 64-bit prefetchable memory, 64-bit non-prefetchable memory.
- BAR 1/3/5: Disabled.

If you select a 64-bit type for BAR 0/2/4, then the adjacent BAR 1/3/5 is disabled.

Table 22. PF BAR Configuration per User Mode [MCDMA, BAM, BAS]

Y = This BAR is available to the user application.

PF0 BAR	Multichannel DMA	Bursting Master	Bursting Slave
BAR0	64-bit prefetchable memory Size: 4 MB - 22 bits *Note:* BAR not available to user application.	Y	Y
BAR1	Disabled *Note:* BAR not available to user application.	Disabled	Disabled
BAR2	64-bit prefetchable memory Size: 4 MB - 22 bits	Y	Y

continued...

PF0 BAR	Multichannel DMA	Bursting Master	Bursting Slave
	Note: BAR not available to user application.		
BAR3	Disabled *Note:* BAR not available to user application.	Disabled	Disabled
BAR4	Disabled *Note:* BAR not available to user application.	Y	Y
BAR5	Disabled *Note:* BAR not available to user application.	Disabled	Disabled
Expansion ROM	Y	Y	Y

Table 23. PF BAR Configuration per User Mode [BAM+BAS, BAM+MCDMA, BAM+BAS +MCDMA]

Y = This BAR is available to the user application.

PF0 BAR	BAM+BAS	BAM+MCDMA	BAM+BAS+MCDMA
BAR0	Y	64-bit prefetchable memory Size: 4 MB - 22 bits *Note:* BAR not available to user application.	64-bit prefetchable memory Size: 4 MB - 22 bits *Note:* BAR not available to user application.
BAR1	Disabled	Disabled *Note:* BAR not available to user application.	Disabled *Note:* BAR not available to user application.
BAR2	Y	Y	Y
BAR3	Disabled	Disabled	Disabled
BAR4	Y	Y	Y
BAR5	Disabled	Disabled	Disabled
Expansion ROM	Y	Y	Y

4.2. Design Examples

Figure 15. Design Example Parameters

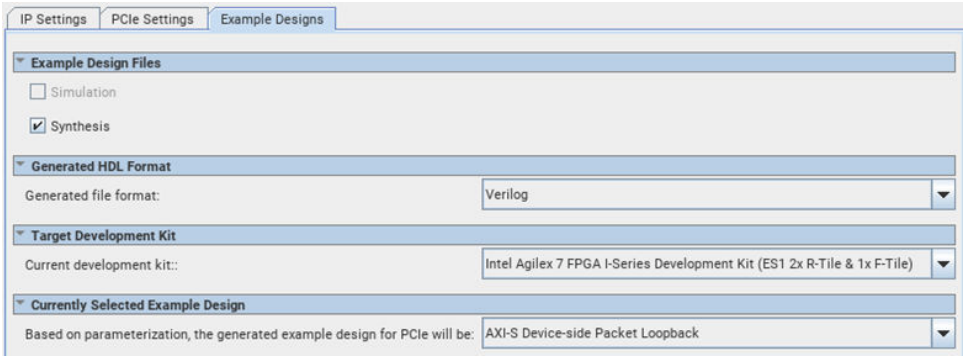


Table 24. Design Example Parameters

Parameter	Value	Default Value	Description
Simulation	**Off**	**Off**	When the **Simulation** box is checked, all necessary filesets for simulation are generated. *Note:* Design example simulation is not supported in the current release.
Synthesis	**On/Off**	**On**	When the **Synthesis** box is checked, all necessary filesets for synthesis are generated. When unchecked, only the Platform Designer system is generated.
Generated file format	**Verilog**	**Verilog**	HDL format.
Current development kit	**None** **Agilex 7 FPGA I-Series Development Kit (ES1 2x R-Tile & 1x F-Tile)** **Agilex 7 FPGA I-Series Development Kit (ES2 2x R-Tile & 1x F-Tile)** **Agilex 7 FPGA I-Series Development Kit (Production 2x R-Tile & 1x F-Tile)**	**Agilex 7 FPGA I-Series Development Kit (ES1 2xR-Tile & 1xF-Tile)**	Selects a target FPGA development kit board. *Note:* If an FPGA development board is selected, the target device used for generation is the one that matches the device on the development kit board.
Currently selected example design	**AXI-MM BAM EP Memory** **AXI-MM Traffic Generator/Checker** **AXI-MM DMA** **AXI-S Device-side Packet Loopback** **AXI-S Packet Generate/Check**	**AXI-S Device-side Packet Loopback**	Selects a design example. *Note:* • Design examples for Root Port mode is not supported. • This parameter needs to be changed based on the **User Mode** and **User Interface** selected in the **MCDMA Settings**. Refer to the table for the design example support for each **User Mode** and **User Interface**. • The AXI-MM DMA design example supports payload sizes only up to 32KB.

Design example generation includes an AXI Streaming PCIe IP (intel_pcie_ss_axi). Hence, the **Example Designs** tab includes the AXI Streaming PCIe IP Parameter Editor. Some of the PCIe Interface parameters are automatically configured per the AXI MCDMA IP parameter values and greyed out.

Figure 16. PCIe Interface Settings Parameters

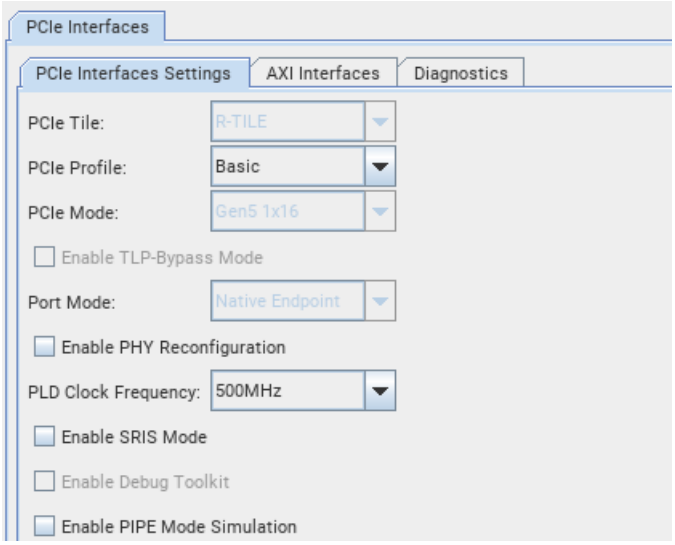


Table 25. PCIe Interface Settings Parameters

Parameter	Value	Default Value	Description
PCIe Tile	**R-Tile**	**R-Tile**	Selects the supported Tile of the PCIe Interface. *Note:* Only R-Tile is supported.
PCIe Profile	**Basic** **Basic+** **Virtual** **Virtual+**	**Basic**	Selects functional features based on the profile like virtualization, additional interfaces, number of endpoints, etc.
PCIe Mode	**Gen5 1x16** **Gen5 2x8** **Gen4 1x16** **Gen4 2x8** **Gen3 1x16** **Gen3 2x8**	**Gen5 1x16**	Selects a PCIe mode. *Note:* For the 2x8 design example, you need to select 1x8 mode in the **IP Settings** > **PCIe Mode**, and then select 2x8 mode in the **Example Designs** > **PCIe Mode** setting.
Port Mode	**Root Port** **Native Endpoint**	**Native Endpoint**	Selects a PCIe mode for the AXI Streaming PCIe IP. *Note:* The mode value is passed from the AXI MCDMA IP Settings. *Note:* The Root Port design example is not supported.
Enable TLP-Bypass Mode	**Off**	**Off**	Enables the TLP-Bypass mode. *Note:* TLP-Bypass mode is not supported in the current release.
Enable PHY Reconfiguration	**On / Off**	**Off**	When on, this parameter creates an Avalon-MM slave interface that software can drive to update Transceiver reconfiguration registers.
PLD Clock Frequency	**Gen5/4 1x16, 2x8: 500/475/450/425/ 400 MHz**	**500 MHz**	Selects the PLD clock frequency.

continued...

Parameter	Value	Default Value	Description
	Gen3 1x16, 2x8: 300/275/250 MHz		
Enable SRIS Mode	**On / Off**	**Off**	Enables a separate reference clock with independent Spread Spectrum Clocking (SSC).
Enable Debug Toolkit	**Off**	**Off**	Debug Toolkit is not supported.
Enable PIPE Mode Simulation	**On / Off**	**Off**	When on, this parameter enables the PIPE mode simulation. *Note:* Design example simulation is not supported in the current release.

Refer to the AXI Streaming IP for PCI Express User Guide for other PCIe Interfaces' parameter descriptions. See Parameter Editor Parameters.

4.2.1. Aligning AXI MCDMA IP Settings with AXI Streaming IP for PCI Express

As the AXI MCDMA IP (intel_pcie_axi_mcdma) needs to interface with the AXI Streaming IP for PCI Express* (intel_pcie_ss_axi), IP settings and interfaces of both IPs must be aligned to ensure both IPs can operate smoothly.

The table below covers IP settings that must be aligned between the two IPs. For example, when the Hard IP Mode of the AXI Streaming IP is configured as a Gen3 x4 Interface 128-bit, the PCIe Mode of the AXI MCDMA IP must be aligned and configured to Gen3 1x4.

Table 26. IP Settings of AXI MCDMA IP and AXI Streaming IP

AXI MCDMA IP Settings		AXI Streaming IP Settings		Description
IP Settings		**PCIe Interface Settings**		
PCIe Mode	Gen5/4/3 1x16	**PCIe Mode**	Gen5/4/3 1x16	
	Gen5/4/3 1x8		Gen5/4/3 1x8 Gen5/4/3 2x8	To generate 1x8, create a quartus.ini file in the project directory with the content: ini_guard=on.
	Gen5/4/3 1x4		Gen5/4/3 1x4 Gen5/4/3 4x4	To generate 1x4, create a quartus.ini file in the project directory with the content: ini_guard=on.
Port Mode	Native Endpoint	**Port Mode**	Native Endpoint	
	Root Port		Root Port	
Multichannel DMA User Mode				
BAR2 Address Width	4 KBytes - 12 bits	**BAR2 Type**	64-bit prefetchable memory	
	...			

continued...

AXI MCDMA IP Settings		AXI Streaming IP Settings		Description
IP Settings		**PCIe Interface Settings**		
	8 EBytes - 63 bits	**BAR2 Size**	4 KBytes - 12 bits	
			...	
			8 EBytes - 63 bits	
Enable multiple physical functions	On / Off	**Enable multiple physical functions**	On / Off	
Total physical functions (PFs)	1 - 8	**Total physical functions (PFs)**	1 - 8	
Enable SR-IOV support	On / Off	**Enable SR-IOV support**	On / Off	
Total virtual functions of physical function	0 - 256	**Total virtual functions of physical function**	0 - 256	
Enable MSI-X	On	**Enable MSI-X**	On	
BAR0 Type	64-bit prefetchable memory	**BAR0 Type**	64-bit prefetchable memory	
BAR0 Size	4 MBytes - 22 bits	**BAR0 Size**	4 MBytes - 22 bits	
BAR1/3/4/5 Type	Disabled	**BAR1/3/4/5 Type**	Disabled	
Expansion ROM Size	Disabled	**Expansion ROM Size**	Disabled	
	...		...	
	16 MBytes - 24 bits		16 MBytes - 24 bits	
BAM+MCDMA, BAM+BAS+MCDMA User Modes				
Enable multiple physical functions	On / Off	**Enable multiple physical functions**	On / Off	
Total physical functions (PFs)	1 - 8	**Total physical functions (PFs)**	1 - 8	
Enable SR-IOV support	On / Off	**Enable SR-IOV support**	On / Off	
Total virtual functions of physical function	0 - 256	**Total virtual functions of physical function**	0 - 256	
Enable MSI-X	On	**Enable MSI-X**	On	
BAR0 Type	64-bit prefetchable memory	**BAR0 Type**	64-bit prefetchable memory	
BAR0 Size	4 MBytes - 22 bits	**BAR0 Size**	4 MBytes - 22 bits	
BAR2 Type	Disabled	**BAR2 Type**	Disabled	
	64-bit prefetchable memory		64-bit prefetchable memory	
	64-bit non-prefetchable memory		64-bit non-prefetchable memory	
BAR2 Size	4 KBytes - 12 bits	**BAR2 Size**	4 KBytes - 12 bits	
	...		...	

continued...

AXI MCDMA IP Settings		AXI Streaming IP Settings		Description
IP Settings		PCIe Interface Settings		
	16 EBytes - 64 bits		16 EBytes - 64 bits	
BAR4 Type	Disabled	**BAR4 Type**	Disabled	BAR4 is needed when BAS is enabled, i.e., in BAM+BAS+MCDMA user mode.
	64-bit prefetchable memory		64-bit prefetchable memory	
	64-bit non-prefetchable memory		64-bit non-prefetchable memory	
BAR4 Size	4 KBytes - 12 bits	**BAR4 Size**	4 KBytes - 12 bits	
	...		...	
	16 EBytes - 64 bits		16 EBytes - 64 bits	
BAR3/5 Type	Disabled	**BAR3/5 Type**	Disabled	
Expansion ROM Size	Disabled	**Expansion ROM Size**	Disabled	
	...		...	
	16 MBytes - 24 bits		16 MBytes - 24 bits	
Endpoint Bursting Master, Bursting Slave, BAM+BAS User Modes				
BAR0/2/4 Type	Disabled	**BAR0/2/4 Type**	Disabled	
	64-bit prefetchable memory		64-bit prefetchable memory	
	64-bit non-prefetchable memory		64-bit non-prefetchable memory	
BAR0/2/4 Size	4 KBytes - 12 bits	**BAR0/2/4 Size**	4 KBytes - 12 bits	
	...		...	
	16 EBytes - 64 bits		16 EBytes - 64 bits	
BAR1/3/5 Type	Disabled	**BAR1/3/5 Type**	Disabled	
Expansion ROM Size	Disabled	**Expansion ROM Size**	Disabled	
	...		...	
	16 MBytes - 24 bits		16 MBytes - 24 bits	
Bursting Slave, BAM+BAS User Modes				
Enable MSI Capability (Native Endpoint mode only)	On / Off	**PF0 Enable MSI**	On / Off	
Enable ATT (Root Port mode only)	On / Off	**Enable Address Translation Services (ATS)**	On / Off	

Some of the AXI MCDMA IP interfaces are enabled by default. This requires the corresponding optional interface of the AXI Streaming IP to be enabled so that signals from both IPs can be connected.

Table 27. Optional Interfaces to be Enabled on the AXI Streaming IP

AXI MCDMA IP Interfaces Enabled by Default	AXI Streaming IP Settings	
Control Shadow Interface (Native Endpoint mode only)	**Enable PCIe0 Control Shadow Interface**	On
Completion Timeout Interface	**Enable PCIe0 Completion Timeout Interface**	On
Configuration Intercept Interface (Native Endpoint mode only)	**Enable PCIe0 Configuration Intercept Interface**	On
Error Interface	**Enable PCIe0 Error Interface**	On

As the `axi_st_clk` and `axi_mm_clk` of the AXI MCDMA IP are expected to be driven by `coreclkout_hip_toapp` of the AXI Streaming IP, the **PLD Clock Frequency** IP parameter must be configured to be aligned with supported frequencies of the AXI MCDMA IP.

Table 28. PLD Clock Frequency Settings

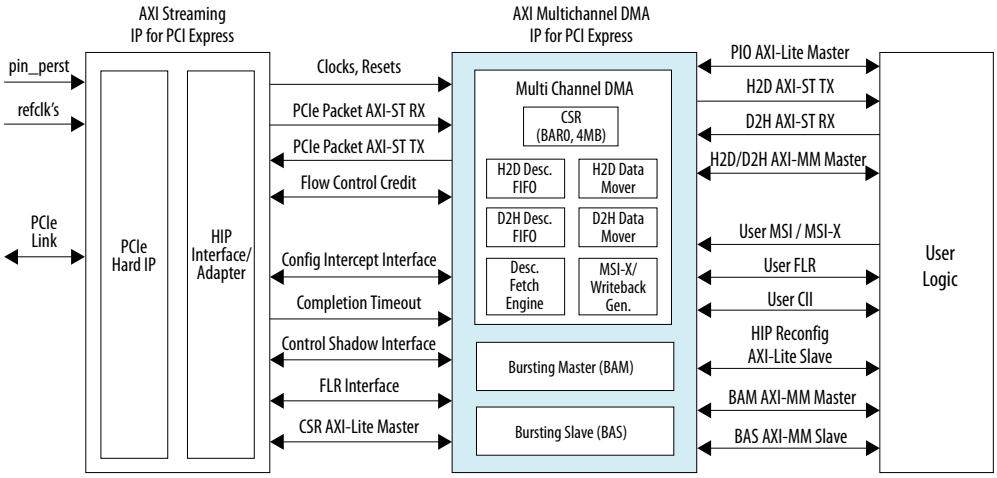
AXI MCDMA IP Supported Clock Frequencies	AXI Streaming IP Settings	
Gen5/4: 400/425/450 MHz	**PLD Clock Frequency**	Gen5/4: 400/425/450 MHz
Gen3: 200/250 MHz		Gen3: 200/250 MHz

5. Interfaces

5.1. System Block Diagram

The following is a system block diagram that shows AXI MCDMA IP interface with the AXI Streaming IP and User Logic.

Figure 17. AXI MCDMA IP System Block Diagram



5.2. Clocks and Resets

Table 29. Clocks

Signal Name	Direction	Description
`axi_st_clk`	Input	Global clock signal for AXI-ST interface. All AXI-ST signals are sampled on the rising edge of this clock. This clock is typically derived from `coreclkout_hip_toapp` output of AXI Streaming IP. Gen5: 500 MHz
`axi_mm_clk`	Input	Global clock signal for AXI-MM interface. All AXI-MM signals are sampled on the rising edge of this clock. This clock is typically derived from `coreclkout_hip_toapp` output of AXI Streaming IP. Gen5: 500 MHz
`axi_lite_clk`	Input	Global clock signal for AXI-Lite interface. All AXI-Lite signals are sampled on the rising edge of this clock. This clock drives control and status register interfaces in the design. Frequency: 100~250 MHz

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*Other names and brands may be claimed as the property of others.

Table 30. Resets

Signal Name	Direction	Description
axi_st_areset_n	Input	Reset signal for AXI Streaming interface
axi_mm_areset_n	Input	Reset signal for AXI-MM interface
axi_lite_areset_n	Input	Reset signal for AXI-Lite interface

5.3. PCIe IP-facing Interfaces

5.3.1. PCIe AXI-ST TX Interface (ss_tx_st)

The outbound packet towards the link is transmitted through this PCIe AXI-ST TX interface with separate header and data interfaces. These interfaces support up to 4 segments of 256-bits with a maximum of 2 Transaction Layer Packets (TLPs) per cycle.

For information on this interface, refer to *Application Logic Guidelines for the AXI Streaming TX Interface in HIP Native Mode (R-Tile)* section in the AXI Streaming IP for PCI Express User Guide.

Interface clock: `axi_st_clk`

NUM_OF_SEG depends on the data width.

NUM_OF_SEG = DWIDTH / SEG_WIDTH

DWIDTH = NUM_OF_SEG * 256

DWIDTH depends on the PCIe mode selected. It is equal to:

- 1024 bits for Gen5 1x16.
- 512 bits for Gen5 1x8, Gen4 1x16, and Gen3 1x16.
- 256 bits for Gen4 1x8, Gen3 1x8, Gen5 1x4, Gen4 1x4, and Gen3 1x4.
- 128 bits for Gen4 1x4 and Gen3 1x4.

Table 31. PCIe AXI-ST TX Interface

Signal Name	Direction	Description
app_ss_st_tx_tvalid	Output	Indicates that the source is driving a valid transfer.
ss_app_st_tx_tready	Input	Indicates that the sink can accept a transfer in the current cycle. *Note:* The **readyLatency** parameter defined in Avalon specification is supported. By default, the value is '0'.
app_ss_st_tx_tdata[DWIDTH-1:0]	Output	Data interface with configurable width specified by DATA_WIDTH parameter. Gen5 1x16 DWIDTH: 4 segments x 256 = 1024
app_ss_st_tx_tkeep[DWIDTH/8-1:0]	Output	A byte qualifier used to indicate whether the content of the associated byte is valid. The invalid bytes are allowed only during app_ss_st_tx_tlast cycle.

continued...

Signal Name	Direction	Description
		Note: The sparse tkeep is not allowed.
app_ss_st_tx_tlast	Output	Indicates End of Data/Command Transmission
app_ss_st_tx_tuser_last_segment [NUM_OF_SEG-1:0]	Output	Indicates the last segment of the packet on tdata bus. For a packet without data payload, this signal is asserted during the header cycle on tuser_hdr bus.
app_ss_st_tx_tuser_hvalid [NUM_OF_SEG-1:0]	Output	Indicates the tuser_hdr is valid in the respective segment.
app_ss_st_tx_tuser_hdr[DWIDTH-1:0]	Output	Carries header format for the respective segment. For the bit positions and mapping, refer to *Header Format* section in the *AXI Streaming IP for PCI Express User Guide*.

5.3.2. PCIe AXI-ST RX Interface (ss_rx_st)

The packets from the link is received on this interface with separate header and data interfaces. These interfaces support up to 4 segments of 256-bits with a maximum of 3 Transaction Layer Packets (TLPs) per cycle.

For information on this interface, refer to the AXI Streaming IP for PCI Express User Guide.

Interface clock: axi_st_clk

NUM_OF_SEG = 4

DWIDTH = NUM_OF_SEG * 256=1024

Table 32. PCIe AXI-ST RX Interface

Signal Name	Direction	Description
ss_app_st_rx_tvalid	Input	Indicates that the source is driving a valid transfer
app_ss_st_rx_tready	Output	Indicates that the sink can accept a transfer in the current cycle. *Note:* The **readyLatency** parameter defined in Avalon specification is supported. By default, the value is '0'.
ss_app_st_rx_tdata[DWIDTH-1:0]	Input	Data interface with configurable width specified by DWIDTH parameter.
ss_app_st_rx_tkeep[DWIDTH/8-1:0]	Input	A byte qualifier used to indicate whether the content of the associated byte is valid. The invalid bytes are allowed only during the ss_app_st_rx_tlast cycle. *Note:* The sparse ss_app_st_rx_tkeep is not allowed.
ss_app_st_rx_tlast	Input	Indicates End of Data/ Command Transmission.

continued...

Signal Name	Direction	Description
		Note: `tlast` may seem redundant with the `tuser_last_segment`. However, tlast can be used by the front-end layer of the AXI BFM/routing fabric that does not deal with the decoding of the multipacket data.
`ss_app_st_rx_tuser_last_segment[NUM_OF_SEG-1:0]`	Input	Indicates Packet End position on tdata bus.
`ss_app_st_rx_tuser_hvalid[NUM_OF_SEG-1:0]`	Input	Indicates the `ss_app_st_rx_tuser_hdr` is valid in the respective segment.
`ss_app_st_rx_tuser_hdr[DWIDTH-1:0]`	Input	`tuser_hdr` is in Power User mode header format

5.3.3. Control and Status Register Interface (ss_csr_lite)

The AXI MCDMA IP uses this interface to access registers implemented in the AXI Streaming IP for PCI Express module, allowing the user application to access PCIe config space registers of all Functions as well as soft register space registers implemented in the AXI Streaming IP.

Application access PCIe config space registers through this interface by adding offset of 0x80000 to the actual physical address of PCIe config space address. Refer to the Register Address Map in the AXI Streaming IP for PCI Express User Guide.

Along with PCIe configuration access, this interface is also used for application error reporting, allowing application to indicate various errors. User Application must write to error generation control registers through this interface to send error information to the PCIe Hard IP block. For more information, refer to the section Application Error Reporting in the AXI Streaming IP for PCI Express User Guide.

The interface follows AXI4-Lite protocol.

Interface clock: `axi_lite_clk`

Table 33. Control and Status Register Interface

Signal Name	Direction	Description
Write Address Channel		
`app_ss_lite_csr_awvalid`	Output	Indicates that the write address channel signals are valid.
`ss_app_lite_csr_awready`	Input	Indicates that a transfer on the write address channel can be accepted.
`app_ss_lite_csr_awaddr[19:0]`	Output	The address of the first transfer in a write transaction.
Write Data Channel		
`app_ss_lite_csr_wvalid`	Output	Indicates that the write data channel signals are valid.
`ss_app_lite_csr_wready`	Input	Indicates that a transfer on the write data channel can be accepted.
`app_ss_lite_csr_wdata[31:0]`	Output	Write data.

continued...

Signal Name	Direction	Description
app_ss_lite_csr_wstrb[3:0]	Output	Write strobe, indicate which byte lanes hold valid data.
Write Response Channel		
ss_app_lite_csr_bvalid	Input	Indicates that the write response channel signals are valid.
app_ss_lite_csr_bready	Output	Indicates that a transfer on the write response channel can be accepted.
ss_app_lite_csr_bresp[1:0]	Input	Write response, indicates the status of a write transaction.
Read Address Channel		
app_ss_lite_csr_arvalid	Output	Indicates that the read address channel signals are valid.
ss_app_lite_csr_arready	Input	Indicates that a transfer on the read address channel can be accepted.
app_ss_lite_csr_araddr[19:0]	Output	The address of the first transfer in a read transaction.
Read Data Channel		
ss_app_lite_csr_rvalid	Input	Indicates that the read data channel signals are valid.
app_ss_lite_csr_rready	Output	Indicates that a transfer on the read data channel can be accepted.
ss_app_lite_csr_rdata[31:0]	Input	Read data.
ss_app_lite_csr_rresp[1:0]	Input	Read response, indicates the status of a read transfer.

5.3.4. Flow Control Credit Interface

5.3.4.1. Transmit Flow Credit Interface (ss_txcrdt)

Before a TLP can be transmitted, flow control logic verifies that the link partner's RX port has sufficient buffer space to accept it. The TX Flow Control interface reports the link partner's available RX buffer space to the MCDMA IP. It reports the space available in units called Flow Control credits for Posted (P), Non-Posted (NP) and Completion (CPL) TLPs (as defined in the RX Flow Control Interface section).

Flow control credits are available for the following TLP categories:

- Posted transactions: TLPs that do not require a response.
- Non-Posted transactions: TLPs that require a completion.
- Completions: TLPs that respond to non-posted transactions.

Interface clock: `axi_st_clk`

Table 34. Transmit Flow Credit Interface

Signal Name	Direction	Description
`ss_app_st_txcrdt_tvalid`	Input	Indicates that the credit information on tdata is valid.
`ss_app_st_txcrdt_tdata[18:0]`	Input	Carries the credit limit information and type of credit. [15:0]: Credit Limit Value • All zero indicates infinite credit for P, NP, CPL [18:16] : Credit Type • 3'b000 - Posted Header Credit • 3'b001 - Non-Posted Header Credit • 3'b010 - Completion Header Credit • 3'b011 - Reserved • 3'b100 - Posted Data Credit • 3'b101 - Non-Posted Data Credit • 3'b110 - Completion Data Credit • 3'b111 - Reserved

5.3.4.2. Receive Flow Control Credit Interface (ss_rxcrdt)

The RX flow control interface provides information on the MCDMA's available RX buffer space for Posted, Non-Posted and Completion transactions to the AXI PCIe IP. It reports the space available in number of credits as specified by the PCIe Specification.

Table 35. Receive Flow Credit Interface

Signal Name	Direction	Description
`app_ss_st_rxcrdt_tvalid`	Output	Indicates that the credit information on tdata is valid.
`app_ss_st_rxcrdt_tdata[18:0]`	Output	Carries the credit limit information and type of credit. [15:0]: Credit Limit Value • All zero indicates infinite credit for P, NP, CPL [18:16]: Credit Type • 3'b000 - Posted Header Credit • 3'b001 - Non-Posted Header Credit • 3'b010 - Completion Header Credit • 3'b011 – Reserved • 3'b100 – Posted Data Credit • 3'b101 - Non-Posted Data Credit • 3'b110 - Completion Data Credit • 3'b111 - Reserved

5.3.5. Configuration Intercept Interface (CII)

The Configuration Intercept Interface (CII) allows the application logic to detect the occurrence of a Configuration (CFG) request on the link. Application monitors this interface to update the PCIe Configuration space register values that are of concern. CII interface gives update only when reconfiguration happens to the writable PCIe configuration space registers. If you want to get the values of the PCIe configuration space registers which are read-only, you must use the HIP reconfiguration interface.

5.3.5.1. Configuration Intercept Request Interface (ss_cii_req)

Interface clock: `axi_lite_clk`

Table 36. Configuration Intercept Request Interface

Signal Name	Direction	Description
ss_app_st_ciireq_tvalid	Input	When asserted, indicates a valid CFG request cycle is waiting to be intercepted. Deasserted when app_ss_st_ciireq_tready is asserted.
app_ss_st_ciireq_tready	Output	Application asserts this signal for one clock to acknowledge ss_app_st_ciireq_tvalid is seen by responder.
ss_app_st_ciireq_tdata[0]	Input	hdr_poisoned: The poisoned bit in the received TLP header on the CII.
ss_app_st_ciireq_tdata[4:1]	Input	hdr_first_be: The first dword byte enable field in the received TLP header on the CII.
ss_app_st_ciireq_tdata[9:5]	Input	slot_num: The slot number in the received TLP header on the CII.
ss_app_st_ciireq_tdata[12:10]	Input	func_num: The PF number in the received TLP header on the CII.
ss_app_st_ciireq_tdata[23:13]	Input	vf_num: The child VF number of parent PF in the received TLP header on the CII.
ss_app_st_ciireq_tdata[24]	Input	vf_active: Indicates VF number is valid in the received TLP header on the CII.
ss_app_st_ciireq_tdata[25]	Input	wr: Indicates a configuration write request detected in the received TLP header on the CII. Also, indicates that ss_app_st_ciireq_tdata[67:36] is valid.
ss_app_st_ciireq_tdata[35:26]	Input	addr: The double word register address in the received TLP header on the CII.
ss_app_st_ciireq_tdata[67:36]	Input	dout: Received TLP payload data from the link partner to your application client. The data is in little endian format. The first received payload byte is in [43:36].
ss_app_st_ciireq_tdata[69:68]	Input	dout: Received TLP payload data from the link partner to your application client. The data is in little endian format. The first received payload byte is in [43:36].
ss_app_st_ciireq_tdata[71:70]	Input	Reserved (SEP Type)

5.3.5.2. Configuration Intercept Response Interface (ss_cii_resp)

The application must return the response for request received on **ss_cii_req** interface using the **ss_cii_resp** interface. The IP is always ready to accept responses from the application. The application provides response data with valid qualifier.

Interface clock : axi_lite_clk

Table 37. Configuration Intercept Response Interface

Signal Name	Direction	Description
`app_ss_st_ciiresp_tvalid`	Output	Application asserts this signal for one clock to indicate that valid data is driven on `app_ss_st_ciiresp_tdata` bus.
`app_ss_st_ciiresp_tdata[31:0]`	Output	Override data from application for the intercepted configuration request on the Configuration Intercept Request interface. For CfgWr: Override the write data to the Configuration register with data supplied by the application logic. For CfgRd: Override the data payload of the completion TLP with data supplied by the application logic.
`app_ss_st_ciiresp_tdata[32]`	Output	Override Data Enable: Application assert this signal to override the CfgWr payload or CfgRd completion using the data supplied by the application logic on `app_ss_st_ciiresp_tdata[31:0]` bus.

5.3.6. Completion Timeout Interface (ss_cplto)

When a completion timeout happens in the AXI Streaming IP, the information is shared with the MCDMA IP through this interface. The interface provides the function number and tag number of the outstanding request that was timed out.

Interface clock: `axi_lite_clk`

Table 38. Completion Timeout Interface

Signal Name	Direction	Description
`ss_app_st_cplto_tvalid`	Input	Indicates that the completion timeout received for outstanding NP request.
`ss_app_st_cplto_tdata[29:0]`	Input	Carries completion Timeout Information. [9:0] - Tag Number [12:10] - PF Number, indicates parent PF number of VF when VF Active is high else PF Number of function. [23:13] - VF Number, indicates VF number when VF Active js high. [24] - VF Active, indicates timeout is for VF. [29:25] - Reserved. All '0's.

5.3.7. Function Level Reset Interface

5.3.7.1. FLR Received Interface (ss_flrrcvd)

Interface clock: axi_lite_clk

Table 39. FLR Received Interface

Signal Name	Direction	Description
ss_app_st_flrrcvd_tvalid	Input	When asserted, indicates a FLR request received from Host. The signal is valid for one clock cycle.
ss_app_st_flrrcvd_tdata[21:0]	Input	[2:0] - The PF number of FLR Request [13:3] - Indicates child VF number of parent PF indicated by PF number. [14] - Indicates request is for Virtual Function implemented in the slot's Physical Function. [19:15] - The slot number of FLR Request [21:20] – The PF number of FLR Request (PF[4:3])

5.3.7.2. FLR Completion Interface (ss_flrcmpl)

Interface clock: axi_lite_clk

Table 40. FLR Completion Interface

Signal Name	Direction	Description
app_ss_st_flrcmpl_tvalid	Output	When asserted, indicates a FLR request completed by MCDMA. The signal is valid for one clock cycle.
app_ss_st_flrcmpl_tdata [21:0]	Output	[2:0] - The PF number of FLR Completion [13:3] - Indicates child VF number of parent PF indicated by PF number. [14] - Indicates completion is from Virtual Function implemented in the slot's Physical Function. [19:15] - The slot number of FLR completion [21:20] – The PF number of FLR Request (PF[4:3])

5.3.8. Control Shadow Interface (ss_ctrlshadow)

Interface clock: axi_lite_clk

Table 41. Control Shadow Interface

Signal Name	Direction	Description
ss_app_st_ctrlshadow_tvalid	Input	AXI MCDMA IP asserts this output for one clock cycle when there is an update to the register fields being monitored due to a Configuration Write performed by the Root Complex. You can copy the new settings of the register fields from the tdata bus.
ss_app_st_ctrlshadow_tdata[39:0]	Input	When app_ctrl_shadow_tvalid has been asserted, this output provides the current settings of the register fields of the associated Function. [2:0] - Identifies the Physical Function Number of configuration register.

continued...

Signal Name	Direction	Description
		[13:3] - Identifies the Virtual Function Number of the configuration register. [14] - Indicates information is for the Virtual Function implemented in the slot's Physical Function. [19:15] - Identifies the slot Number of the configuration register. [20] - Bus Master Enable [21] - MSI-X Mask [22] - MSI-X Enable [23] - Mem Space Enable [24] - ExpRom Enable [25] - TPH Req Enable [26] - ATS Enable [27] - MSI Enable [28] - MSI Mask [29] - Extended Tag [30] - 10 Bit Tag Req Enable [31] - PTM Enable [34:32] - MPS Size [37:35] - MRRS Size [38] - VF Enable [39] - Page Request Enable

5.4. User-facing Interfaces

5.4.1. H2D AXI-ST Source (h2d_st_initatr)

The H2D AXI source interface is used to send H2D DMA data to the external AXI sink logic.

Interface clock: axi_st_clk

Table 42. H2D AXI-ST Source Interface

Signal Name	Direction	Description
`h2d_axi_st_tvalid`	Output	Indicates that the source is driving a valid transfer.
`h2d_axi_st_tready`	Input	Indicates that the sink can accept a transfer in the current cycle. *Note:* The readyLatency parameter defined in Avalon specification is supported. By default the value is '0'.
`h2d_axi_st_tdata[1023:0]`	Output	Data interface
`h2d_axi_st_tkeep[127:0]`	Output	A byte qualifier used to indicate whether the content of the associated byte is valid. The invalid bytes are allowed only during h2d_axi_st_tlast cycle. The sparse tkeep is not allowed.
`h2d_axi_st_tlast`	Output	Indicates end of data transmission.

continued...

Signal Name	Direction	Description
h2d_axi_st_tid[11:0]	Output	Stream ID, indicates channel number.
h2d_axi_st_tuser_metadata[63:0]	Output	Descriptor 8-byte metadata. Available only when metadata support is enabled through the IP Parameter Editor.
h2d_axi_st_tuser_error	Output	Error Status

5.4.2. D2H AXI-ST Sink (d2h_st_respndr)

The D2H AXI Sink interface is used to sink D2H DMA data from the external AXI source logic.

Interface clock: axi_st_clk

Table 43. D2H AXI-ST Sink Interface

Signal Name	Direction	Description
d2h_axi_st_tvalid	Input	Indicates that the source is driving a valid transfer
d2h_axi_st_tready	Output	Indicates that the sink can accept a transfer in the current cycle. *Note:* The readyLatency parameter defined in Avalon specification is supported. By default the value is '0'.
d2h_axi_st_tdata[1023:0]	Input	Data interface
d2h_axi_st_tkeep[127:0]	Input	A byte qualifier used to indicate whether the content of the associated byte is valid. The invalid bytes are allowed only during d2h_axi_st_tlast cycle. The sparse tkeep is not allowed.
d2h_axi_st_tlast	Input	Indicates end of data transmission
d2h_axi_st_tid[11:0]	Input	Stream ID, indicates channel number
d2h_axi_st_tuser_metadata[63:0]	Input	Descriptor 8-byte metadata. Available only when metadata support is enabled through the IP Parameter Editor.

5.4.3. H2D/D2H AXI-MM Master

The MCDMA mode supports an AXI Memory-Mapped (AXI-MM) interface which can be selected in the IP Parameter Editor. The AXI-MM master interface consists of a single AXI interface, where the Write Channel is connected to the H2D DMA path and the Read Channel is connected to the D2H DMA path.

5.4.3.1. AXI-MM Write (H2D)

The AXI-MM Write channel is used to transfer H2D DMA data to the external AXI-MM responder. This interface supports a write width of up to 1024 bits, enabling a maximum of 512 bytes per AXI-MM transaction.

The ReadyLatency of this port is enabled and set as follows:

- 64 for a 128-bit data width
- 32 for a 256-bit data width
- 16 for a 512-bit data width
- 8 for a 1024-bit data width

This ReadyLatency allows the master to continue transferring data even after the ready signal has been de-asserted.

Table 44. AXI-MM Write Channel (H2D)

Signal Name	Direction	Description
Write Address Channel		
dma_axi_mm_awvalid	Output	Write address valid. This signal indicates that the channel is signaling valid write address and control information.
dma_axi_mm_awready	Input	Write address ready. This signal indicates that the slave is ready to accept an address and associated control signals.
dma_axi_mm_awid[3:0]	Output	Transaction ID for the Write channel. Depends on the number of outstanding requests (default is 4 outstanding requests).
dma_axi_mm_awaddr[63:0]	Output	Write address. The write address gives the address of the first transfer in a write burst transaction.
dma_axi_mm_awlen[7:0]	Output	Burst length. The burst length gives the exact number of transfers in a burst. This information determines the number of data transfers associated with the address.
dma_axi_mm_awsize[2:0]	Output	Burst size. This signal indicates the size of each transfer in the burst.
dma_axi_mm_awburst[1:0]	Output	Burst type. The burst type, along with the burst size information, determines how the address is calculated for each transfer within the burst. AXI MCDMA supports only INCR burst type.
dma_axi_mm_awprot[2:0]	Output	Protection type. This signal indicates the privilege and security level of the transaction, and whether the transaction is a data access or an instruction access. (not used)
dma_axi_mm_awlock	Output	Lock type. Tied to '0'.
Write Data Channel		
dma_axi_mm_wvalid	Output	Write valid.
dma_axi_mm_wready	Input	Write ready. Indicates the receiver can accept write data.
dma_axi_mm_wdata x16: [1023:0] x8: [511:0] x4: [255:0]	Output	Write data. The data width depends on the PCIe mode configuration.
dma_axi_mm_wstrb x16: [127:0] x8: [63:0] x4: [31:0]	Output	Write strobes. Signal width: data width / 8.
dma_axi_mm_wlast	Output	Write last.

continued...

Signal Name	Direction	Description
Write Response Channel		
dma_axi_mm_bvalid	Input	Write response valid
dma_axi_mm_bready	Output	Write response ready
dma_axi_mm_bid[3:0]	Input	Transaction ID for the Write channel.
dma_axi_mm_bresp[1:0]	Input	Write response.

5.4.3.2. AXI-MM Read Channel (D2H)

The AXI-MM Read channel is used to read D2H DMA data from the external AXI-MM responder. This port is capable of reading a maximum of 512 bytes of data per AXI-MM transaction.

Table 45. AXI-MM Read Channel (D2H)

Signal Name	Direction	Description
Read Address Channel		
dma_axi_mm_arvalid	Output	Read address valid. This signal indicates that the channel is signaling valid read address and control information.
dma_axi_mm_arready	Input	Read address ready. This signal indicates that the slave is ready to accept an address and associated control signals.
dma_axi_mm_arid[3:0]	Output	Transaction ID for the Read. Depends on the number of outstanding requests (default is 4 outstanding requests).
dma_axi_mm_araddr[63:0]	Output	Read address. The read address gives the address of the first transfer in a read burst transaction.
dma_axi_mm_arlen[7:0]	Output	Burst length. The burst length gives the exact number of transfers in a burst. This information determines the number of data transfers associated with the address.
dma_axi_mm_arsize[2:0]	Output	Burst size. This signal indicates the size of each transfer in the burst.
dma_axi_mm_arburst[1:0]	Output	Burst type. The burst type, along with the burst size information, determines how the address is calculated for each transfer within the burst. AXI MCDMA supports only INCR burst type.
dma_axi_mm_arprot[2:0]	Output	Protection type. This signal indicates the privilege and security level of the transaction, and whether the transaction is a data access or an instruction access (not used).
dma_axi_mm_arlock	Output	Lock type. Tied to '0'.
Read Data Channel		
dma_axi_mm_rvalid	Input	Read valid. This signal indicates that the channel is signaling the required read data.
dma_axi_mm_rready	Output	Read ready. This signal indicates that the master can accept the read data and response information.
dma_axi_mm_rlast	Input	Read last. This signal indicates the last transfer in a read burst.

continued...

Signal Name	Direction	Description
dma_axi_mm_rid[3:0]	Input	Read ID tag. Depends on number of outstanding requests. Default is 4 outstanding.
dma_axi_mm_rdata x16: [1023:0] x8: [511:0] x4: [255:0]	Input	Read data. Data bus width varies based on the PCIe link width.
dma_axi_mm_rresp	Input	Read response. This signal indicates the status of the read transfer. EXOKAY is not supported.

5.4.4. BAM AXI-MM Master (bam_mm_initatr)

The BAM bypasses the AXI MCDMA IP for PCI Express and provides a way for a Host to perform bursting PIO read/writes to the user logic. The BAM converts memory read and write TLPs initiated by the remote link partner and received over the PCIe link into AXI-MM burst read and write transactions, and sends back CplD TLPs for read requests it receives.

The BAM interface address mapping is as follows:

BAM address = {vf_active, pf[PF_NUM_W-1:0], vf[VF_NUM_W-1:0], bar_num[2:0], address}

1. **vf_active**: This indicates that SRIOV is enabled.
2. **pf[PF_NUM_W-1:0]**: This denotes the physical function number decoded from the PCIe header. PF_NUM_W, which is defined as ($clog2(Number of PFs)) is a parameter in the RTL design chosen to limit the number of wires on the user interface of the Multichannel DMA IP. This reduces the overhead associated with unnecessary wiring.
3. **vf[VF_NUM_W-1:0]**: This denotes the virtual function number decoded from the PCIe header. VF_NUM_W, which is defined as ($clog2(Number of VFs)) is a parameter in the RTL design chosen to limit the number of wires on the user interface of the Multichannel DMA IP. This reduces the overhead associated with unnecessary wiring.
4. **bar_num[2:0]**: This denotes the BAR number decoded from the PCIe header.
5. **address**: Number of bits required for BAR2 size requested across all Functions (PFs and VFs). Example: If BAR2 is selected as 4 MB, the address size is 22 bits.

Interface Clock: `axi_mm_clk`

DWIDTH:

- 1024 bits for Gen5 1x16.
- 512 bits for Gen5 1x8, Gen4 1x16, and Gen3 1x16.
- 256 bits for Gen4 1x8, Gen3 1x8, Gen5 1x4, Gen4 1x4, and Gen3 1x4.
- 128 bits for Gen4 1x4 and Gen3 1x4.

Note: For the Root Port: In the Root Port mode, the AXI-MM address output from BAM is the same as the one received on the PCIe AXI-ST RX Interface (st_rx).

Table 46. BAM AXI-MM Master Interface

Signal Name	Direction	Description
Write Address Channel		
bam_axi_mm_awvalid	Output	Write address valid. This signal indicates that the channel is signaling valid write address and control information.
bam_axi_mm_awready	Input	Write address ready. This signal indicates that the slave is ready to accept an address and associated control signals.
bam_axi_mm_awid[3:0]	Output	Write address ID. This signal is the identification tag for the write address group of signals. The default value is 0.
bam_axi_mm_awaddr[n:0]	Output	Write address. The write address gives the address of the first transfer in a write burst transaction. Refer to the BAM address mapping for the actual address width.
bam_axi_mm_awlen[7:0]	Output	Burst length. The burst length gives the exact number of transfers in a burst. This information determines the number of data transfers associated with the address.
bam_axi_mm_awsize[2:0]	Output	Burst size. This signal indicates the size of each transfer in the burst.
bam_axi_mm_awburst[1:0]	Output	Burst type. The burst type and the size information determine how the address for each transfer within the burst is calculated. The FPGA supports only the INCR burst type.
bam_axi_mm_awlock	Output	Lock type. Tied to '0'.
bam_axi_mm_awprot[2:0]	Output	Protection type. This signal indicates the privilege and security level of the transaction, and whether the transaction is a data access or an instruction access. (not used)
Write Data Channel		
bam_axi_mm_wvalid	Output	Write Data Valid
bam_axi_mm_wready	Input	Write Data Ready Indicates the receiver can accept write data
bam_axi_mm_wdata[DWIDTH-1:0]	Output	Write data
bam_axi_mm_wstrb[DWIDTH/8-1:0]	Output	Write strobes
bam_axi_mm_wlast	Output	Write last
Write Response Channel		
bam_axi_mm_bvalid	Input	Write Response Valid
bam_axi_mm_bready	Output	Write Response Ready

continued...

Signal Name	Direction	Description
bam_axi_mm_bid[3:0]	Input	Response ID This signal is the ID tag of the write response.
bam_axi_mm_bresp[1:0]	Input	Write Response
Read Address Channel		
bam_axi_mm_arvalid	Output	Read address valid. This signal indicates that the channel is signaling valid read address and control information.
bam_axi_mm_arready	Input	Read address ready. This signal indicates that the slave is ready to accept an address and associated control signals
bam_axi_mm_arid[3:0]	Output	Read address ID. This signal is the identification tag for the read address group of signals. The default value is 0.
bam_axi_mm_araddr[n:0]	Output	Read address. The read address gives the address of the first transfer in a read burst transaction. Refer to the BAM address mapping for the actual address width.
bam_axi_mm_arlen[7:0]	Output	Burst length. The burst length gives the exact number of transfers in a burst.
bam_axi_mm_arsize[2:0]	Output	Burst size. This signal indicates the size of each transfer in the burst.
bam_axi_mm_arburst[1:0]	Output	Burst type. The burst type and the size information determine how the address for each transfer within the burst is calculated. The FPGA supports only the INCR burst type.
bam_axi_mm_arlock	Output	Lock type. Tied to '0'.
bam_axi_mm_arprot[2:0]	Output	Protection type. This signal indicates the privilege and security level of the transaction, and whether the transaction is a data access or an instruction access. (not used)
Read Data Channel		
bam_axi_mm_rvalid	Input	Read data valid. This signal indicates that the channel is signaling the read data is valid.
bam_axi_mm_rready	Output	Read data ready. This signal indicates that the master can accept the read data and response information.
bam_axi_mm_rid[3:0]	Input	Read ID tag. This signal is the identification tag for the read data group of signals generated by the slave.
bam_axi_mm_rdata[DWIDTH-1:0]	Input	Read Data
bam_axi_mm_rresp[1:0]	Input	Read response. This signal indicates the status of the read transfer. EXOKAY is not supported on the FPGA. (Per the AXI specification, EXOKAY means the Exclusive read succeeded. This response is only permitted for an Exclusive read).

continued...

Signal Name	Direction	Description
		• 2'b00: OKAY • 2'b01: EXOKAY • 2'b10: SLVERR • 2'b11: DECERR
bam_axi_mm_rlast	Input	Read last. This signal indicates the last transfer in a read burst.

5.4.5. BAS AXI-MM Slave (bas_mm_respndr)

The AXI-MM Bursting Slave module translates AXI-MM Read address channel and Write address channel transactions from the user logic to PCI Express Mrd and Mwr TLPs respectively. The returned PCI Express CplD packets are translated to AXI-MM Read Data channel interface as a response to the AXI-MM read address channel transaction.

Interface Clock: axi_mm_clk

DWIDTH:

- 1024 bits for Gen5 1x16.
- 512 bits for Gen5 1x8, Gen4 1x16, and Gen3 1x16.
- 256 bits for Gen4 1x8, Gen3 1x8, Gen5 1x4, Gen4 1x4, and Gen3 1x4.
- 128 bits for Gen4 1x4 and Gen3 1x4.

Table 47. BAS AXI-MM Slave Interface

Signal Name	Direction	Description
Write Address Channel		
bas_axi_mm_awvalid	Input	Write address valid. This signal indicates that the channel is signaling valid write address and control information.
bas_axi_mm_awready	Output	Write address ready. This signal indicates that the slave is ready to accept an address and associated control signals.
bas_axi_mm_awid[3:0]	Input	Write address ID. This signal is the identification tag for the write address group of signals.
bas_axi_mm_awaddr[63:0]	Input	Write address. The write address gives the address of the first transfer in a write burst transaction. The Bursting Slave's addresses must be aligned to the width of the data bus. For example, if the data width is 64B, the addresses must align to 64B.
bas_axi_mm_awuser[15:0]	Input	Write Address User. The write address user information encodes the following information: • EP mode: User-defined signal - {1'b0, vf_num[11:0], vf_active, pf_num[2:0]} • RP mode: User-defined signal - to be used as the Requester ID in the Memory Write TLPs.

continued...

Signal Name	Direction	Description
bas_axi_mm_awlen[7:0]	Input	Burst length. The burst length gives the exact number of transfers in a burst. This information determines the number of data transfers associated with the address.
bas_axi_mm_awsize[2:0]	Input	Burst size. This signal indicates the size of each transfer in the burst.
bas_axi_mm_awburst[1:0]	Input	Burst type. The burst type and the size information, determine how the address for each transfer within the burst is calculated. Only INCR burst type is supported.
bas_axi_mm_awlock	Input	Lock type. Tied to '0'.
bas_axi_mm_awuser[15:0]	Input	Root Port mode: User-defined signal - to be used as the Requester ID in the TLPs. Endpoint mode: User-defined signal - {vf_num[11:0], vf_active, pf_num[2:0]}.
bas_axi_mm_awprot[2:0]	Input	Protection type. This signal indicates the privilege and security level of the transaction, and whether the transaction is a data access or an instruction access. (not used)
Write Data Channel		
bas_axi_mm_wvalid	Input	Write Data Valid
bas_axi_mm_wready	Output	Write Data Ready Indicates the receiver can accept write data
bas_axi_mm_wdata[DWIDTH-1:0]	Input	Write Data
bas_axi_mm_wstrb[DWIDTH/8-1:0]	Input	Write Strobes
bas_axi_mm_wlast	Input	Write Last
Write Response Channel		
bas_axi_mm_bvalid	Output	Write Response Valid
bas_axi_mm_bready	Input	Write Response Ready
bas_axi_mm_bid[3:0]	Output	Response ID. This signal is the identification tag of the write response.
bas_axi_mm_bresp[1:0]	Output	Write Response
Read Address Channel		
bas_axi_mm_arvalid	Input	Read address valid. This signal indicates that the channel is signaling valid read address and control information.
bas_axi_mm_arready	Output	Read address ready. This signal indicates that the slave is ready to accept an address and associated control signals
bas_axi_mm_arid[3:0]	Input	Read address ID.

continued...

Signal Name	Direction	Description
		This signal is the identification tag for the read address group of signals.
bas_axi_mm_araddr[63:0]	Input	Read address. The read address gives the address of the first transfer in a read burst transaction. The Bursting Slave's addresses must be aligned to the width of the data bus. For example, if the data width is 64B, the addresses must align to 64B.
bas_axi_mm_aruser[15:0]	Input	Read address user. The read address user signal encodes the following information: • EP mode: User-defined signal - {1'b0, vf_num[11:0], vf_active, pf_num[2:0]} • RP mode: User-defined signal - to be used as the Requester ID in the Memory Read TLPs.
bas_axi_mm_arlen[7:0]	Input	Burst length. The burst length gives the exact number of transfers in a burst.
bas_axi_mm_arsize[2:0]	Input	Burst size. This signal indicates the size of each transfer in the burst.
bas_axi_mm_arburst[1:0]	Input	Burst type. The burst type and the size information, determine how the address for each transfer within the burst is calculated. Only INCR burst type is supported.
bas_axi_mm_arlock	Input	Lock type. Tied to '0'.
bas_axi_mm_aruser[15:0]	Input	User-defined signal - {vf_num[11:0], vf_active, pf_num[2:0]}
bas_axi_mm_arprot[2:0]	Input	Protection type. This signal indicates the privilege and security level of the transaction, and whether the transaction is a data access or an instruction access. (not used)
Read Data Channel		
bas_axi_mm_rvalid	Output	Read data valid. This signal indicates that the channel is signaling the required read data.
bas_axi_mm_rready	Input	Read data ready. This signal indicates that the master can accept the read data and response information.
bas_axi_mm_rid[3:0]	Output	Read ID tag. This signal is the identification tag for the read data group of signals generated by the slave.

continued...

Signal Name	Direction	Description
bas_axi_mm_rdata[DWIDTH-1:0]	Output	Read Data
bas_axi_mm_rresp[1:0]	Output	Read response. This signal indicates the status of the read transfer. EXOKAY is not supported on Agilex 7 FPGAs.
bas_axi_mm_rlast	Output	Read last. This signal indicates the last transfer in a read burst.

5.4.6. PIO AXI-Lite Master (pio_lite_initiatr)

The PIO Master is enabled in Multichannel DMA mode. It bypasses the DMA block and provides a way for the Host to do MMIO read/write to CSR registers of user logic. PCIe BAR2 is mapped to the PIO Master. Any TLP targeting BAR2 is forwarded to the user logic via AXI-Lite interface. TLP address targeting the PIO interface should be 8 bytes aligned. The PIO interface supports non-bursting 64-bit write and read transfers only.

Note: Do not attempt to perform 32-bit transactions on PIO interface. Only 64-bit transactions are supported.

The PIO interface address mapping is as follows:

```
PIO address = {vf_active, pf[PF_NUM_W-1:0], vf[VF_NUM_W-1:0], address}
```

1. **vf_active:** This indicates that SRIOV is enabled
2. **pf[PF_NUM_W-1:0]:** This denotes the physical function number decoded from the PCIe header. PF_NUM_W, which is defined as ($clog2(Number of PFs)) is a parameter in the RTL design chosen to limit the number of wires on the user interface of the Multichannel DMA IP. This reduces the overhead associated with unnecessary wiring.
3. **vf[VF_NUM_W-1:0]:** This denotes the virtual function number decoded from the PCIe header. VF_NUM_W, which is defined as ($clog2(Number of VFs)) is a parameter in the RTL design chosen to limit the number of wires on the user interface of the Multichannel DMA IP. This reduces the overhead associated with unnecessary wiring.
4. **address:** Number of bits required for BAR2 size requested across all Functions (PFs and VFs). Example: If BAR2 is selected as 4 MB, the address size is 22 bits.

The following table describes the interface signals.

Interface clock: axi_lite_clk

Table 48. PIO AXI-Lite Master Interface

Signal Name	Direction	Description
Write Address Channel		
rx_pio_axi_lite_awvalid	Output	Write address valid
rx_pio_axi_lite_awready	Input	Write address ready Indicates that the slave is ready to accept an address.
rx_pio_axi_lite_awaddr[n:0]	Output	Write address.

continued...

Signal Name	Direction	Description
		The write address gives the address of the first transfer in a write burst transaction. Refer to the PIO address mapping for actual address width.
rx_pio_axi_lite_awprot[2:0]	Output	Protection type. This signal indicates the privilege and security level of the transaction, and whether the transaction is a data access or an instruction access. (not used)
Write Data Channel		
rx_pio_axi_lite_wvalid	Output	Write data valid
rx_pio_axi_lite_wready	Input	Write data ready Indicates that the slave is ready to accept data
rx_pio_axi_lite_wdata[63:0]	Output	Write data
rx_pio_axi_lite_wstrb[7:0]	Output	Write strobe Indicates which byte lanes hold valid data
Write Response Channel		
rx_pio_axi_lite_bvalid	Input	Indicates that the write response channel signals are valid
rx_pio_axi_lite_bready	Output	Indicates that a transfer on the write response channel can be accepted
rx_pio_axi_lite_bresp[1:0]	Input	Write response Indicates the status of a write transaction
Read Address Channel		
rx_pio_axi_lite_arvalid	Output	Indicates that the read address channel signals are valid
rx_pio_axi_lite_arready	Input	Indicates that a transfer on the read address channel can be accepted
rx_pio_axi_lite_araddr[n:0]	Output	The address of the first transfer in a read transaction. Refer to the PIO address mapping for actual address width.
rx_pio_axi_lite_arprot[2:0]	Output	Protection type. This signal indicates the privilege and security level of the transaction, and whether the transaction is a data access or an instruction access. (not used)
Read Data Channel		
rx_pio_axi_lite_rvalid	Input	Read data valid
rx_pio_axi_lite_rready	Output	Indicates the master is ready to accept read data
rx_pio_axi_lite_rdata[63:0]	Input	Read data
rx_pio_axi_lite_rresp[1:0]	Input	Read response, indicates the status of the read transfer. EXOKAY is not supported on the FPGA.

5.4.7. HIP Reconfig AXI-Lite Slave (user_csr_lite)

The Hard IP Reconfiguration interface is an AXI-Lite slave interface with a 21-bit address and a 32-bit data bus. You can use this bus to dynamically modify the value of configuration registers that are read-only at run time.

Note: After a warm reset or cold reset, changes made to the configuration registers of the Hard IP via the Hard IP reconfiguration interface are lost as these registers revert back to their default values.

Interface clock: `axi_lite_clk`

Table 49. HIP Reconfig AXI-Lite Slave Interface

Signal Name	Direction	Description
Write Address Channel		
`usr_hip_reconfig_awvalid`	Input	Indicates that the write address channel signals are valid
`usr_hip_reconfig_awready`	Output	Indicates that a transfer on the write address channel can be accepted
`usr_hip_reconfig_awaddr[19:0]`	Input	The address of the first transfer in a write transaction
Write Data Channel		
`usr_hip_reconfig_wvalid`	Input	Indicates that the write data channel signals are valid
`usr_hip_reconfig_wready`	Output	Indicates that a transfer on the write data channel can be accepted
`usr_hip_reconfig_wdata[31:0]`	Input	Write data
`usr_hip_reconfig_wstrb[3:0]`	Input	Write strobes Indicates which byte lanes hold valid data
Write Response Channel		
`usr_hip_reconfig_bvalid`	Output	Indicates that the write response channel signals are valid
`usr_hip_reconfig_bready`	Input	Indicates that a transfer on the write response channel can be accepted
`usr_hip_reconfig_bresp[1:0]`	Output	Write response Indicates the status of a write transaction
Read Address Channel		
`usr_hip_reconfig_arvalid`	Input	Indicates that the read address channel signals are valid
`usr_hip_reconfig_arready`	Output	Indicates that a transfer on the read address channel can be accepted
`usr_hip_reconfig_araddr[19:0]`	Input	The address of the first transfer in a read transaction
Read Data Channel		
`usr_hip_reconfig_rvalid`	Output	Indicates that the read data channel signals are valid

continued...

Signal Name	Direction	Description
usr_hip_reconfig_rready	Input	Indicates that a transfer on the read data channel can be accepted
usr_hip_reconfig_rdata[31:0]	Output	Read data
usr_hip_reconfig_rresp[1:0]	Output	Read response Indicates the status of a read transfer. • 2'b00: OKAY • 2'b01: EXOKAY • 2'b10: SLVERR • 2'b11: DECERR

5.4.8. User Event MSI-X (user_msix)

User logic can request the DMA engine to send an event interrupt for a queue associated with a PF/VF.

Interface clock : axi_lite_clk

Table 50. User Event MSI-X Interface

Signal Name	Direction	Description
user_event_msix_tvalid	Input	The valid signal qualifies valid data on any cycle with data transfer.
user_event_msix_tready	Output	On interfaces supporting backpressure, the sink asserts ready to mark the cycles where transfers may take place.
user_event_msix_tdata[15:0]	Input	{msix_queue_dir, rsvd[3:0], msix_queue_num_i[10:0]} *Note:* msix_queue_dir denotes DMA direction. • 0: Device-to-Host (D2H) DMA Queue • 1: Host-to-Device (H2D) DMA Queue

5.4.9. User Event MSI (user_msi)

Interface clock : axi_lite_clk

Table 51. User Event MSI Interface

Signal Name	Direction	Description
user_msi_req	Input	MSI request signal. Assertion causes MemWrite TLP to be generated to Host based on the MSI Capability register values and other MSI input ports. You can deassert MSI request any time after Acknowledge (user_msi_ack) has been asserted.
user_msi_ack	Output	Acknowledge for MSI request. Asserted for 1 clock cycle. User logic can deassert the MSI request as soon as this signal is asserted.
user_msi_num[4:0]	Input	MSI number that indicates the offset between the base message data and the MSI to send.

continued...

Signal Name	Direction	Description
		When multiple message mode is enabled, this signal sets the lower five bits of the MSI Data register. `user_msi_num[4:0]` is not an option in the IP Parameter Editor. The user application provides the MSI vector information when multiple messages are enabled. For more information, refer to the Message Data Register for MSI in the PCIe base specification.
`user_msi_func_num`	Input	Specifies the function number requesting an MSI transmission
`user_msi_status[1:0]`	Output	Indicates the execution status of requested MSI. Valid when `user_msi_ack` is asserted. • 00: MSI message sent • 01: Pending bit is set and no message sent (MSI is masked) • 10: Error (MSI is not enabled or not allocated) • 11: Reserved

5.4.10. User Function Level Reset (user_flr)

When the DMA engine receives Functional Level Resets from Host, the reset requests are propagated to the downstream logic via this interface. In addition to performing resets to its internal logic, the FLR interface waits for an acknowledgment from user logic for the reset request before it issues an acknowledgement to the PCIe Hard IP.

Interface clock : `axi_lite_clk`

Table 52. User Function Level Reset Interface

Signal Name	Direction	Description
`user_flr_rcvd_val`	Output	Indicates user logic to begin FLR for the specified channel in `usr_flr_rcvd_chan_num`. Remains asserted until `usr_flr_completed` input is sampled 1'b1.
`user_flr_rcvd_chan_num[11:0]`	Output	Indicates channel number for which FLR has to be initiated by user logic
`user_flr_completed`	Input	One-cycle pulse from user logic indicates completion of FLR activity for channel in `usr_flr_rcvd_chan_num`

5.4.11. User Configuration Intercept Interface - EP Only

The User Configuration Intercept Interface (CII) allows the application logic to detect the occurrence of a Configuration (CFG) request on the link and to modify its behavior. The application logic can use the CII to:

- Delay the processing of a CFG request by the controller. This allows the application to perform any housekeeping task first.
- Overwrite the data payload of a CfgWr request. The application logic can also overwrite the data payload of a CfgRd Completion TLP.

5.4.11.1. User Configuration Intercept Request Interface (user_cii_req)

Interface clock: `axi_lite_clk`

Table 53. User Configuration Intercept Request Interface (user_cii_req)

Signal Name	Direction	Description
`dma_user_st_ciireq_tvalid`	Output	When asserted, indicates a valid CFG request cycle is waiting to be intercepted. Deasserted when `user_dma_st_ciireq_tready` is asserted.
`user_dma_st_ciireq_tready`	Input	Application asserts this signal for one clock to acknowledge that `dma_user_st_ciireq_tvalid` is seen by the responder.
`dma_user_st_ciireq_tdata[71:0]`	Output	Refer to Configuration Intercept Request Interface for the `ss_app_st_ciireq_tdata[71:0]` signal description.

5.4.11.2. User Configuration Intercept Response Interface (user_cii_resp)

Interface clock: `axi_lite_clk`

Table 54. User Configuration Intercept Response Interface (user_cii_resp)

Signal Name	Direction	Description
`user_dma_st_ciiresp_tvalid`	Input	Application asserts this signal for one clock to indicate that valid data is driven on the dma_user_st_ciiresp_tdata bus.
`dma_user_st_ciiresp_tdata[31:0]`	Input	Override data. • CfgWr: this bus contains the data supplied by the application logic to override the write data to the PCIe Hard IP register. • CfgRd: this bus contains the data supplied by the application logic to override the data payload of the Completion TLP.
`dma_user_st_ciiresp_tdata[32]`	Input	Override Data Enable. Application assert this signal to override the CfgWr payload or CfgRd completion using the data supplied by the application logic on the `dma_user_st_ciiresp_tdata[31:0]` bus. .

5.4.12. Configuration Slave (cs_lite_respndr) - RP Only

The Configuration Slave (CS) interface is applicable only in Root Port mode. For functional description, refer to Configuration Slave (CS) (14-bit address format) in the Multi Channel DMA IP for PCI Express User Guide. Note: The AVMM interface is replaced with AXI Lite.

Interface clock: `axi_lite_clk`

Table 55. Configuration Slave Interface

Signal Name	Direction	Description
Write Address Channel		
`cs_axi_lite_awvalid`	Input	Write address valid.

continued...

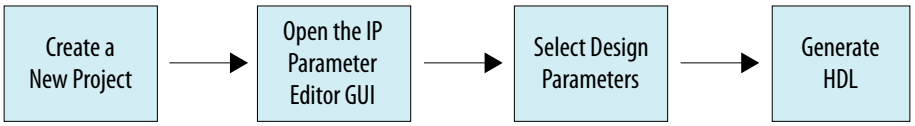
Signal Name	Direction	Description
cs_axi_lite_awready	Output	Write address ready.
cs_axi_lite_awaddr[13:0]	Input	Write address. .
Write Data Channel		
cs_axi_lite_wvalid	Input	Write data valid.
cs_axi_lite_wready	Output	Write data ready.
cs_axi_lite_wdata[31:0]	Input	Write data.
cs_axi_lite_wstrb[3:0]	Input	Write data strobes. This indicates which byte lanes are valid.
Write Response Channel		
cs_axi_lite_bvalid	Output	Write response valid.
cs_axi_lite_bready	Input	Write response ready.
cs_axi_lite_bresp[1:0]	Output	Write response. This signal indicates the status of the write transaction.
Read Address Channel		
cs_axi_lite_arvalid	Input	Read address valid.
cs_axi_lite_arready	Output	Read address ready.
cs_axi_lite_araddr[13:0]	Input	Read address.
Read Data Channel		
cs_axi_lite_rvalid	Output	Read data valid.
cs_axi_lite_rready	Input	Read data ready.
cs_axi_lite_rdata[31:0]	Output	Read data.
cs_axi_lite_rresp[1:0]	Output	Read response. This signal indicates the status of the read transfer. EXOKAY is not supported on the FPGA.

6. Designing with the IP Core

6.1. Generating the IP Core

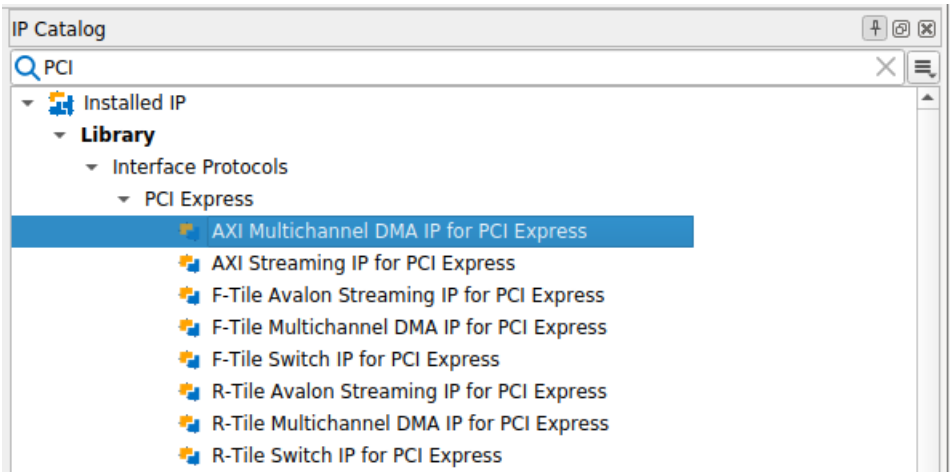
You can use the Quartus Prime Pro Edition IP Catalog or Platform Designer to define and generate an AXI Multichannel DMA IP for PCI Express custom component. Follow the steps shown in the figure below to generate a custom AXI Multichannel DMA IP for PCI Express component.

Figure 18. IP Generation Flowchart



You can select AXI Multichannel DMA IP for PCI Express in the Quartus Prime Pro Edition IP Catalog or Platform Designer as shown below.

Figure 19. Quartus Prime Pro Edition IP Catalog (with filter applied)



Open the IP Parameter Editor GUI and select the design parameters. Select **Generate HDL...** to generate the HDL files for synthesis and/or simulation.

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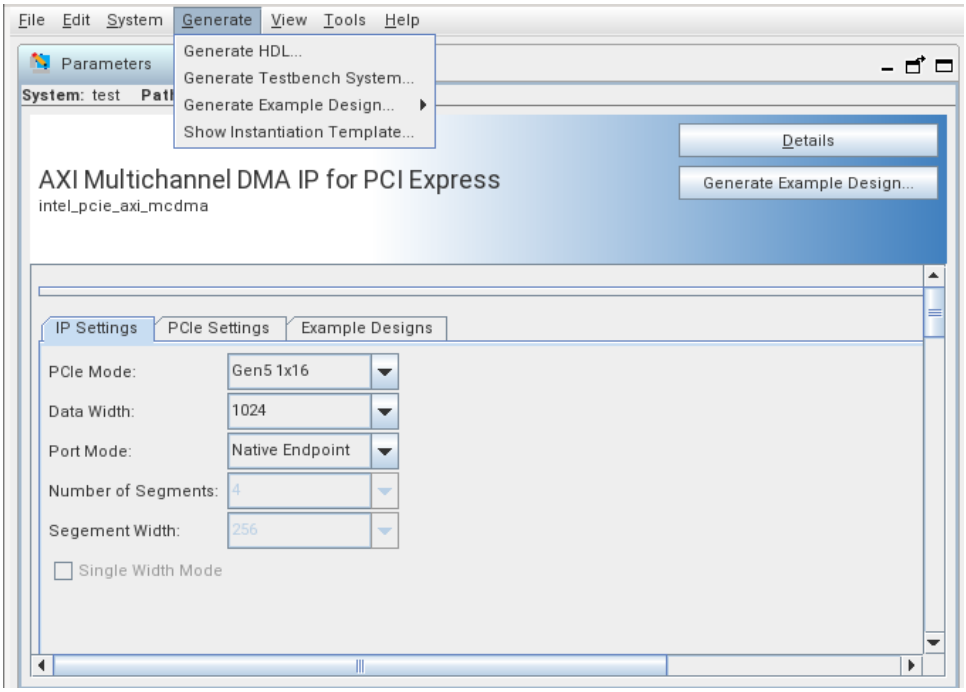
*Other names and brands may be claimed as the property of others.

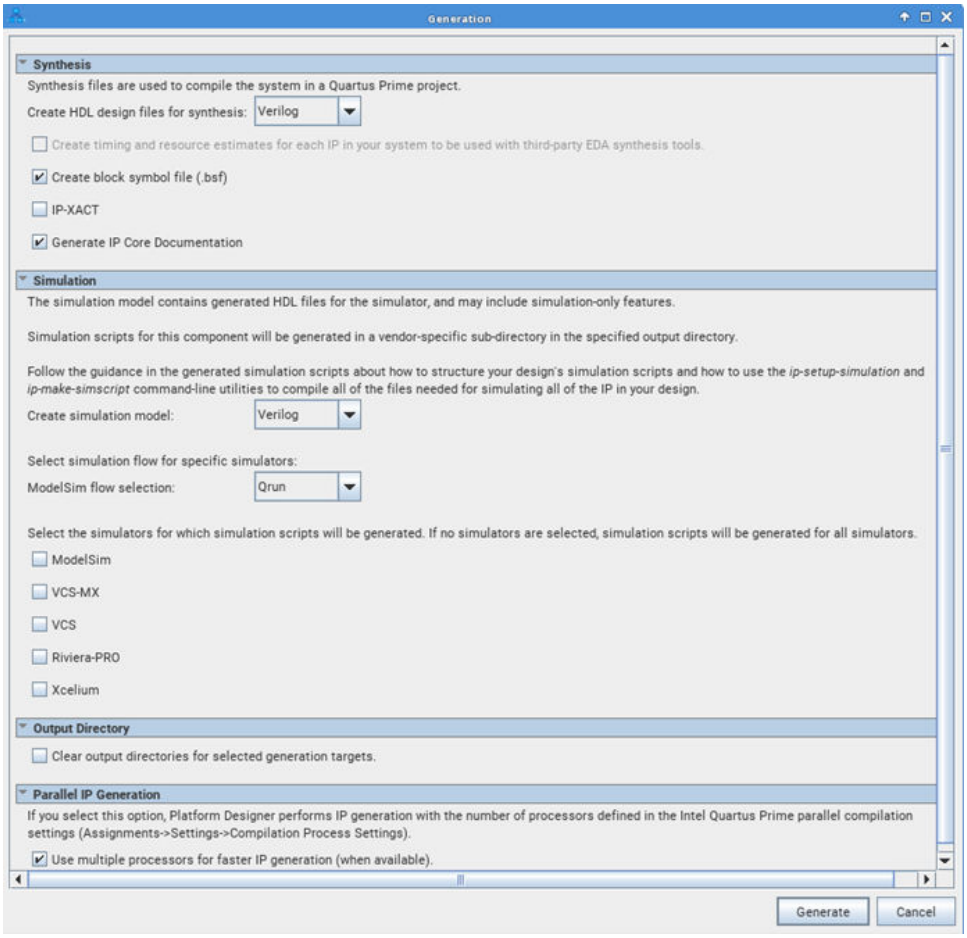
Figure 20. Select Design Parameters and Generate HDL in the IP Parameter Editor GUI



To enable IP simulation model generation, set **Create simulation model** to **Verilog** or **VHDL** when you generate HDL.

Also, select the simulators for which simulation scripts are generated. If no simulators are selected, simulation scripts are generated for all simulators.

Figure 21. Generate HDL Options for Synthesis and Simulation



6.2. IP Core Generation Output

The Quartus Prime Pro Edition software generates the following output file structure for individual IP cores that are not part of a Platform Designer system.

Figure 22. IP Core Generation Output

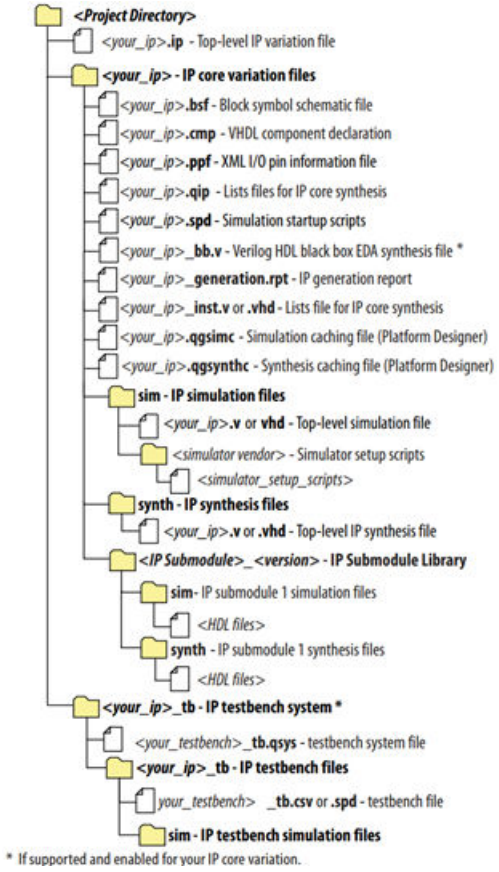


Table 56. Output Files of IP Generation

File Name	Description
`<your_ip>.ip`	Top-level IP variation file that contains the parameterization of an IP core in your project. If the IP variation is part of a Platform Designer system, the parameter editor also generates a .qsys file.
`<your_ip>.cmp`	The VHDL Component Declaration (.cmp) file is a text file that contains local generic and port definitions that you use in VHDL design files.
`<your_ip>_generation.rpt`	IP or Platform Designer generation log file. Displays a summary of the messages during IP generation.
`<your_ip>.qgsimc` (Platform Designer systems only)	Simulation caching file that compares the .qsys and .ip files with the current parameterization of the Platform Designer system and IP core. This comparison determines if Platform Designer can skip regeneration of the HDL
`<your_ip>.qgsynthc` (Platform Designer systems only)	Synthesis caching file that compares the .qsys and .ip files with the current parameterization of the Platform Designer system and IP core.

continued...

File Name	Description
	This comparison determines if Platform Designer can skip regeneration of the HDL.
`<your_ip>.qip`	Contains all information to integrate and compile the IP component.
`<your_ip>.csv`	Contains information about the upgrade status of the IP component.
`<your_ip>.bsf`	A symbol representation of the IP variation for use in Block Diagram Files (.bdf).
`<your_ip>.spd`	Input file that ip-make-simscript requires to generate simulation scripts. The .spd file contains a list of files you generate for simulation, along with information about memories that you initialize.
`<your_ip>.ppf`	The Pin Planner File (.ppf) stores the port and node assignments for IP components you create for use with the Pin Planner.
`<your_ip>_bb.v`	Use the Verilog blackbox (_bb.v) file as an empty module declaration for use as a blackbox.
`<your_ip>_inst.v` or `<your_ip>_inst.vhd`	HDL example instantiation template. Copy and paste the contents of this file into your HDL file to instantiate the IP variation.
`<your_ip>.regmap`	If the IP contains register information, the Quartus Prime Software generates the .regmap file. The .regmap file describes the register map information of master and slave interfaces. This file complements the .sopcinfo file by providing more detailed register information about the system. This file enables register display views and user customizable statistics in System Console.
`<your_ip>.svd`	Allows HPS System Debug tools to view the register maps of peripherals that connect to HPS within a Platform Designer system. During synthesis, the Quartus Prime Software stores the .svd files for slave interface visible to the System Console masters in the .sof file in the debug session. System Console reads this section, which Platform Designer queries for register map information. For system slaves, Platform Designer accesses the registers by name.
`<your_ip>_inst.v` `<your_ip>_inst.vhd`	HDL files that instantiate each submodule or child IP core for synthesis or simulation.
`sim/mentor/`	Contains msim_setup.tcl and run_msim_setup.tcl scripts to set up and run a ModelSim® simulation.
`sim/aldec/`	Contains Riviera-PRO* scripts rivierapro_setup.tcl and run_rivierapro_setup.tcl to setup and run a simulation.
`sim/synopsys/vcs/` `sim/synopsys/vcsmx/`	Contains a shell script vcs_setup.sh to set up and run a VCS* simulation. Contains a shell script vcsmx_setup.sh and synopsys_sim.setup file to set up and run a VCS MX simulation.

continued...

File Name	Description
sim/xcelium/	Contains a shell script xcelium_setup.sh and other setup files to set up and run an Xcelium* simulation.
/submodules/	Contains HDL files for the IP core submodule.
/<IP submodule>/	Platform Designer generates /synth and /sim sub-directories for each IP submodule directory that Platform Designer generates.

6.3. Simulating the IP Core

The Quartus Prime Pro Edition Software optionally generates a functional simulation model of the IP core and vendor-specific simulator setup scripts when you generate your parameterized AXI Multichannel DMA IP core.

Note: Simulating the IP core with Quartus Prime design example testbench is not supported in the current release.

6.4. Quick Start Guide

6.4.1. Design Example Overview

The AXI Multi Channel DMA IP for PCI Express design examples demonstrate a multi-channel DMA solution for Agilex 7 devices using the soft IP implemented in the FPGA fabric.

You can generate the design examples from the **Example Designs** tab of the AXI Multi Channel DMA IP for PCI Express Parameter Editor. The tables below summarize the user interface, user mode, drivers, application and tests supported by each design example:

- AXI-MM DMA
- AXI-MM Traffic Generator/Checker
- AXI-MM BAM EP Memory
- AXI-S Device side Packet Loopback
- AXI-S Packet Generate/Check

Note: The DPDK driver supports up to 256 DMA channels.

Table 57. AXI-MM DMA Design Example

MCDMA User Interface: AXI-MM

User Mode	Driver	Application and Supported Test
Multichannel DMA	Custom	perfq_app: • PIO test • AXI-MM DMA test
	DPDK	mcdma_test: • PIO test • AXI-MM DMA test
BAM + MCDMA	Custom	perfq_app:

continued...

User Mode	Driver	Application and Supported Test
		• AXI-MM DMA test • BAM test
	DPDK	mcdma_test: • AXI-MM DMA test • BAM test
BAM + BAS + MCDMA	Custom	perfq_app: • AXI-MM DMA test • BAM test • BAS test
	DPDK	mcdma_test: • AXI-MM DMA test • BAM test • BAS test

Table 58. AXI-MM Traffic Generator/Checker Design Example

User Mode	Driver	Application and Supported Test
BAM + BAS	Custom	perfq_app: • BAM test • BAS test
	DPDK	mcdma_test: • BAM test • BAS test

Table 59. AXI-MM BAM EP Memory Design Example

User Mode	Driver	Application and Supported Test
Bursting Master	Custom	perfq_app: • BAM test
	DPDK	mcdma_test: • BAM test

Table 60. AXI-S Device-side Packet Loopback Design Example

MCDMA User Interface: AXI-S

User Mode	Driver	Application and Supported Test
Multichannel DMA	Custom	perfq_app: • PIO test • AXI-ST Packet Loopback DMA test
	DPDK	mcdma_test: • PIO test • AXI-ST Packet Loopback DMA test
BAM + MCDMA	Custom	perfq_app: • AXI-ST Packet Loopback DMA test • BAM test
	DPDK	mcdma_test:

continued...

User Mode	Driver	Application and Supported Test
		• AXI-ST Packet Loopback DMA test • BAM test
BAM + BAS + MCDMA	Custom	perfq_app: • AXI-ST Packet Loopback DMA test • BAM test • BAS test
	DPDK	mcdma_test: • AXI-ST Packet Loopback DMA test • BAM test • BAS test

Table 61. AXI-S Packet Generate/Check Design Example

MCDMA User Interface: AXI-S

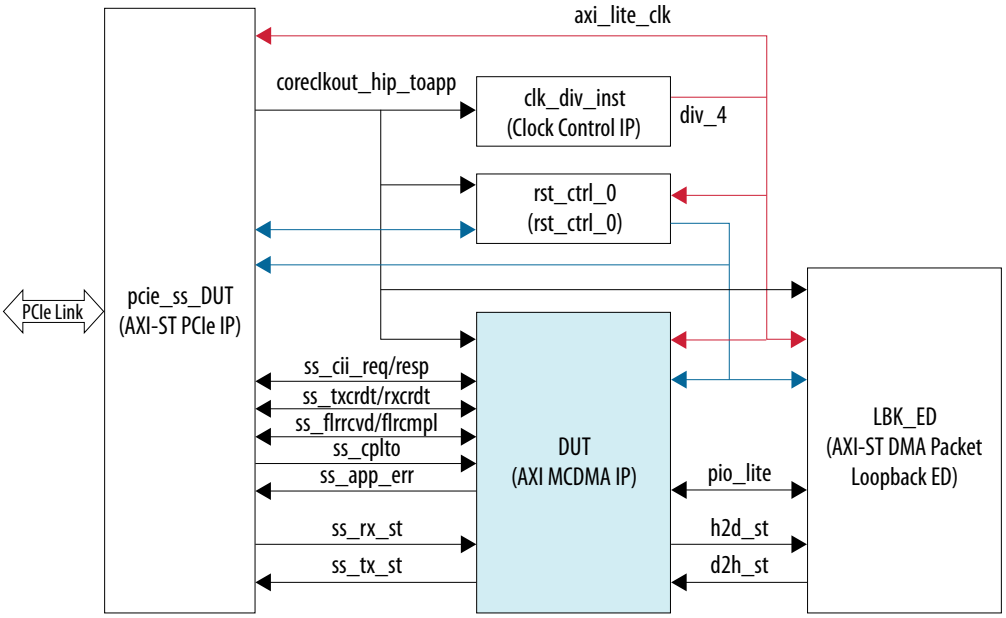
User Mode	Driver	Application and Supported Test
Multichannel DMA	Custom	perfq_app: • PIO test • AXI-ST Packet Generate/Check DMA test
	DPDK	mcdma_test: • PIO test • AXI-ST Packet Generate/Check DMA test

6.4.2. Functional Description

In this device-side packet loopback design example, the Host initially sets up specific memory locations within its memory. Data from the Host memory is then transferred to the device-side memory by the AXI Multichannel DMA for PCI Express IP through Host-to-Device (H2D) DMA operations. Subsequently, the IP loops this data back to the Host memory using Device-to-Host (D2H) DMA operations.

Additionally, the design example enables the AXI-Lite PIO master, which bypasses the DMA controller. This allows the application running in the Host to perform single, non-bursting register read/write operations with the on-chip memory block.

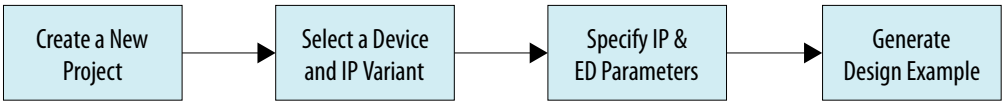
Figure 23. AXI Streaming Device-side Packet Loopback Design Example



6.4.3. Generating the Design Example

Using Quartus Prime, you can generate a design example for the AXI Multichannel DMA IP.

Figure 24. Design Example Generation



Following is the procedure to generate a design example:

1. In the Quartus Prime Pro Edition software, create a new project by clicking **File** > **New Project Wizard**. Click **Next**.
2. Specify the **Directory, Name, and Top-Level Entity**. Click **Next**.
3. Specify the **Family, Device & Board Settings** as follows:
   a. Select **Family**: Agilex 7 (F-Series/M-Series/I-Series).
   b. Select **Target device**: **Filter (Name)**: AGIB027R29A1E2VB.
   c. Click the target device in the table.

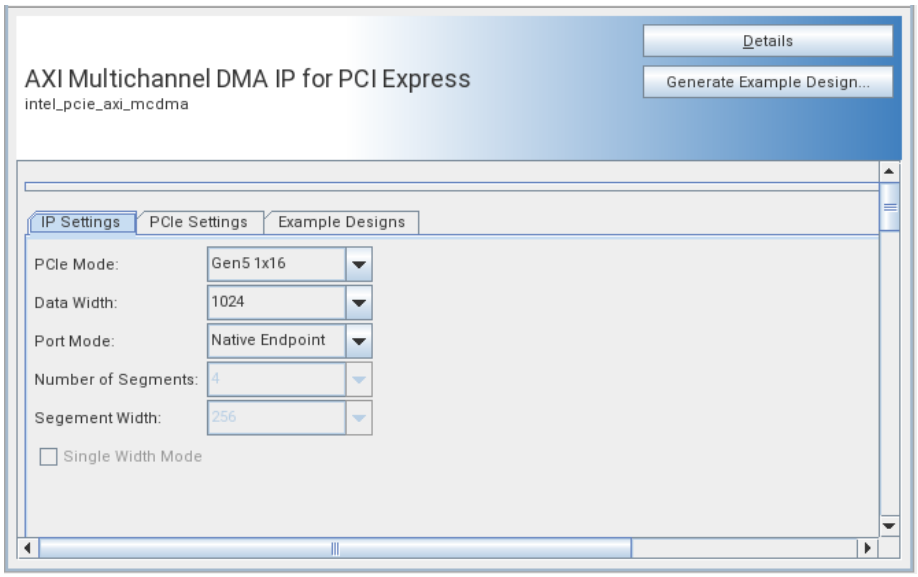
   Note: The device used for the Quartus Prime-generated design example matches the device used in the target FPGA development kit board and may not be the same as the target device selected here.

Figure 25. Selecting Device Family and Target Device

d. Click **Finish**.

4. In the **IP Catalog** pane, type PCI in the Filter, and select **AXI Multichannel DMA IP for PCI Express** under Interface Protocols > PCI Express. Click **+ Add...**.
5. In the **New IP Variant** dialogue box, specify the file name. Click **Create**.
6. Configure the IP parameters if needed in the **IP Settings** and **PCIe Settings** tabs.

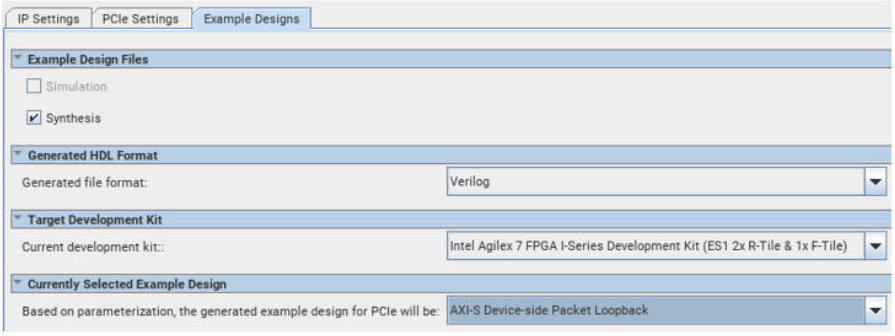
Figure 26. IP Parameter Editor



7. Go to the **Example Designs** tab. Select the **Target Development Kit** and **Currently Selected Example Design**.

   Note: The current release does not support design example simulation.

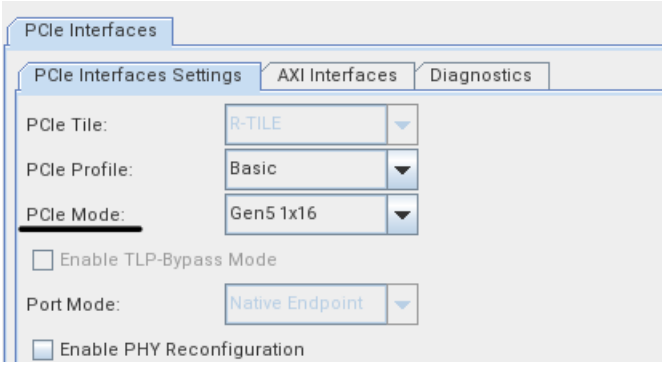
Figure 27. Design Example Tab



8. Go to **PCIe Interfaces** > **PCIe Interface Settings** > **PCIe Mode** and select a PCIe Hard IP mode.

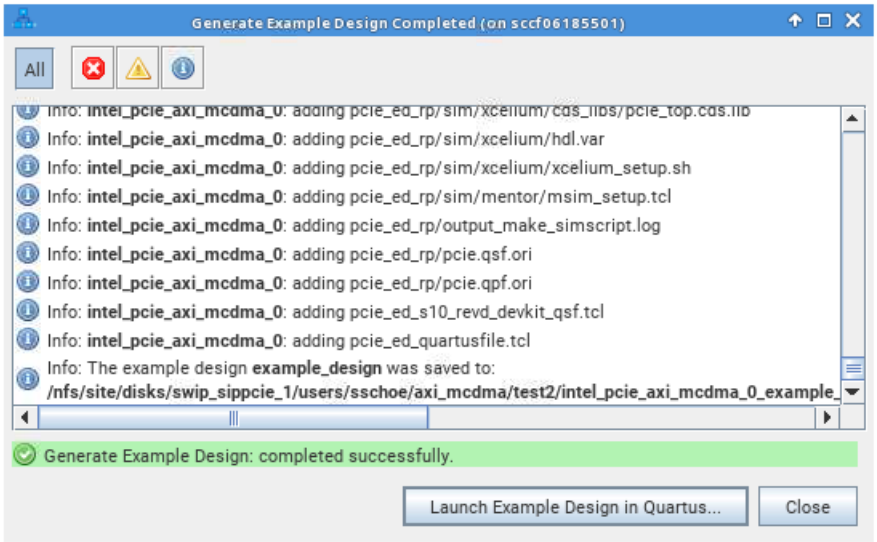
   Note: 1x8, 1x4, and 4x4 design examples are not supported in the current release.

Figure 28. PCIe Mode Selection



9. Click **Generate Example Design**. After the design example generation is completed, click **Launch Example Design in Quartus**.

Figure 29. Launch Design Example



10. In Quartus Prime, click **Processing** > **Start Compilation** to compile the design example and generate the bitstream (.sof) file.

    Note: The .sof file and all other Quartus Prime compilation reports (for example, the STA report, synthesis report, fitter report) are generated in the `output_files` folder in the project directory.

6.4.4. Simulating the Design Example

Design example simulation is not supported in the current release.

6.4.5. Testing the Design Example

For information about software test setup (OS/Kernel/gcc version) and test application, refer to this Quick Start Guide section.

7. Known Issues

Table 62. Known Issues

Issue	HSD-ES	Found In	Status
When you enable address byte aligned transfers, you may observe a DMA failure for lower payload sizes on the AXI interface.	16023190098	24.1	Byte aligned access is not supported. Address should be aligned to the data width. A fix is planned for a future release.
A Virtual Function DMA with MSIX enabled gets stuck.	16024369928	24.2	A fix is planned for a future release.
You may observe the AXI-MM DMA being stuck, and a Queue reset failure, during a hardware test.	16024808887	24.3	Fixed in 25.1.1.
A BAS data validation failure is observed in BAM+BAS+MCDMA mode using the AXI-MM interface during a hardware test.	16026624756	25.1	A fix is planned for a future release.

IP upgrades from v24.1 to v25.1, and from v24.2 to v25.1, will fail.	16027079112	25.1	To upgrade the IP from 1.0.0 to 3.0.0 (Quartus Prime 25.1), add `app_err` interface signals in the top-level file which is `<design example>/pcie_ed/synth/pcie_ed.v`:
			`wire dut_ss_app_err_tvalid;`
			`wire dut_ss_app_err_tready;`
			`wire [31:0] dut_ss_app_err_tdata;`
			`wire dut_ss_app_err_tlast;`
			`wire [13:0] dut_app_ss_st_err_tuser_error_type_app_ss_st_err_tuser_error_type;`

Rx(D2H) data validation error.	16027730547	25.1.1	A fix is planned for a future release.
Rx Stuck and Queue reset failure with DMA test.	16027528472	25.1.1	A fix is planned for a future release.
X propagation in the header with ATT enabled and address not aligned to data width.	16027824074	25.1.1	Fixed in 25.3.
In bidirectional DMA test using AXI-S Packet Generate/Check design example, performance may drop with increasing payload size/channel.	16027711385	25.1.1	A fix is planned for a future release.
H2D DMA performance drop with AXI-S Packet Generate/Check design example.	16027711269	25.1.1	A fix is planned for a future release.
User MSI Interrupt gets response of error code = 0x02	22021404440	25.1	Fixed in 25.3.1.

continued...

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Issue	HSD-ES	Found In	Status
Arsize is not properly handled for BAM read requests.	16028282145	25.3	Fixed in 25.3.1.
The maximum outstanding non-posted requests supported by the IP is less than 32 with BAM mode.	16027239939	25.1.1	Fixed in 25.3.1. The BAM interface can support up to 32 non-posted requests.
Fails to compile with ModelSim. Error: Variable 'tlp_msgd_s2' written by continuous and procedural assignments.	18043399015	25.1.1	Fixed in 25.3.1.
Gen5x16 Multichannel DMA in AXI-MM mode fails with the queues being stuck when the link is downgraded from Gen5 to Gen4.	18044053451	25.3	A fix is planned for a future release.
Intermittent performance drops are observed with the Packet Generate/ Check design example in Gen5x16 when the number of DMA channels is greater than 16.	16027711385	25.1.1	A fix is planned for a future release.

8. Revision History for the AXI Multichannel DMA IP for PCI Express* User Guide

Document Version	Quartus Prime Version	IP Version	Changes
2026.01.12	25.3.1	3.3.0	Added a couple of items to the *Known Issues* section.
2026.01.07	25.3.1	3.3.0	• Added a Note about payload size support (up to 32KB) to *Design Examples*. • Changed the IP name from 'AXI Multichannel DMA FPGA IP for PCI Express' to 'AXI Multichannel DMA IP for PCI Express' throughout the document. • Updated the list in the *Known Issues* section.
2025.08.11	25.1.1	3.1.0	• Added the AXI-S Packet Generate/Check design example to the table in *Design Example Support*. • Updated the file names and descriptions in the table in *IP Core Generation Output*. • Added the *Design Example Overview* section under *Quick Start Guide*. • Updated the table in *Known Issues*.
2025.04.07	25.1	3.0.0	• Added an H2D/D2H AXI-MM interface to *Endpoint Mode Features*. • Added the section *Design Example Support*. • Updated the block diagram in *Functional Description*. • Updated the system block diagram in *Interfaces*. • Added the section *H2D/D2H AXI-MM Master* to the *User-facing Interfaces* section. • Updated the table in *Known Issues*.
2025.01.24	24.3.1	2.2.0	• Updated the parameter descriptions in *IP Settings*. • Updated the screenshots and descriptions for various modes under *MCDMA Settings*. • Updated the steps in *Generating the Design Example*. • Added more items to *Known Issues*.
2024.11.25	24.3	2.1.0	Added bug fixes.
2024.07.22	24.2	2.0.0	• Added more links to related documents to the *Related Information* section. • Added the *IP and Design Example Support* section. • Added the *Quick Start Guide* section.
2024.05.23	24.1	1.0.0	Initial Release

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